
2026 Fall Linear Algebraic Groups Note

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Contents

1	Definition and basic properties	4
1.1	Definitions and conventions	4
1.1.1	Algebraic groups as functors	7
1.2	Basic properties	11
1.3	The image of a homomorphism	24
1.4	Group actions	25
1.5	Structure of general algebraic groups	28
2	Linear representations of linear algebraic groups	31
2.1	Hopf algebras	31
2.2	Linear representations	35
2.3	Comodules	36
2.4	LAG is linear	39
2.5	Constructing all finite-dimensional representations	42
A	Homework	45
A.1	Week 1	45
A.2	Week 2	56

Introduction

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Goal. Understand the structure of linear algebraic groups (LAG) over a field k .

Remark. Faithfully follow Miaofen Chen's lecture in Fudan University, 2026 Fall.

1 Definition and basic properties

1.1 Definitions and conventions

Let k be a field. For a scheme X of finite type over k , let

$$X^{\text{cl}} := \{\text{closed points of } X\}, \quad j : X^{\text{cl}} \hookrightarrow X, \quad \mathcal{O}_{X^{\text{cl}}} := j^{-1}\mathcal{O}_X.$$

The locally ringed spaces obtained in this way are the *ultra-schemes* of [Gro66, §10.9] For an affine scheme,

$$(\text{Spec } A)^{\text{cl}} = \text{Spm } A,$$

the maximal spectrum of A .

Proposition 1.1. [Gro66, 10.9.6] The functor $X \mapsto X^{\text{cl}}$, with the induced structure sheaf, gives an equivalence between the category of Jacobson schemes and the category of ultra-schemes.

Convention. From now on, schemes over k are schemes of finite type over k . Under the preceding equivalence, affine schemes are written as $\text{Spm } A$. Thus a scheme over k admits a finite affine covering

$$X = \bigcup_{i=1}^r \text{Spm } A_i.$$

Definition 1.2. An *algebraic group over k* is a scheme G/k together with morphisms

$$m : G \times_k G \longrightarrow G, \quad e : \text{Spec } k \longrightarrow G, \quad i : G \longrightarrow G,$$

called multiplication, unit, and inverse, satisfying the following properties.

(1) **Associativity.**

$$\begin{array}{ccc} G \times G \times G & \xrightarrow{m \times \text{id}_G} & G \times G \\ \text{id}_G \times m \downarrow & & \downarrow m \\ G \times G & \xrightarrow{m} & G \end{array}$$

(2) **Existence of a unit.** Writing $p_G : G \rightarrow \text{Spec } k$ for the structure morphism,

$$m \circ (\text{id}_G, e \circ p_G) = \text{id}_G = m \circ (e \circ p_G, \text{id}_G).$$

Equivalently, the two diagrams

$$\begin{array}{ccc}
 G \simeq G \times \operatorname{Spec} k & \xrightarrow{\operatorname{id}_G \times e} & G \times G \\
 & \searrow \operatorname{id}_G & \downarrow m \\
 & & G
 \end{array}
 \qquad
 \begin{array}{ccc}
 G \simeq \operatorname{Spec} k \times G & \xrightarrow{e \times \operatorname{id}_G} & G \times G \\
 & \searrow \operatorname{id}_G & \downarrow m \\
 & & G
 \end{array}$$

commute.

(3) Inverse property.

$$m \circ (\operatorname{id}_G, i) = e \circ p_G = m \circ (i, \operatorname{id}_G).$$

Equivalently,

$$\begin{array}{ccccc}
 & & \operatorname{Spec} k & & \\
 & \nearrow p_G & & \searrow e & \\
 G & \xrightarrow{(\operatorname{id}_G, i)} & G \times G & \xrightarrow{m} & G \\
 & \searrow (i, \operatorname{id}_G) & & \nearrow m & \\
 & & G \times G & &
 \end{array}$$

Definition 1.3. A *linear algebraic group over k* is an algebraic group over k whose underlying scheme is affine.

Definition 1.4. An algebraic group G is *commutative* if the diagram

$$\begin{array}{ccc}
 G \times G & \xrightarrow{\tau} & G \times G \\
 & \searrow m & \downarrow m \\
 & & G
 \end{array}
 \qquad (\tau(x, y) = (y, x))$$

commutes.

Remark. Let $G = \operatorname{Spm} A$, where A is a finitely generated commutative k -algebra. The structure morphisms of G correspond contravariantly to

$$m^\# : A \longrightarrow A \otimes_k A, \quad e^\# : A \longrightarrow k, \quad i^\# : A \longrightarrow A.$$

These are called, respectively, the *comultiplication*, *counit*, and *coinverse* (or *antipode*).

Moreover,

$$A \simeq \text{Hom}_{k\text{-schemes}}(G, \mathbb{A}_k^1).$$

If

$$m^\#(f) = \sum_j f_j \otimes g_j,$$

then for $x, y \in G$,

$$f(m(x, y)) = \sum_j f_j(x)g_j(y).$$

The group axioms translate into the following identities for $m^\#, e^\#, i^\#$.

Coassociativity.

$$(m^\# \otimes \text{id}_A) \circ m^\# = (\text{id}_A \otimes m^\#) \circ m^\#.$$

Equivalently,

$$\begin{array}{ccc} A \otimes A \otimes A & \xleftarrow{\text{id}_A \otimes m^\#} & A \otimes A \\ \uparrow m^\# \otimes \text{id}_A & & \uparrow m^\# \\ A \otimes A & \xleftarrow{m^\#} & A \end{array}$$

Counit. Using $k \otimes A \simeq A \simeq A \otimes k$,

$$(e^\# \otimes \text{id}_A) \circ m^\# = \text{id}_A = (\text{id}_A \otimes e^\#) \circ m^\#.$$

Equivalently,

$$\begin{array}{ccc} A & \xleftarrow{\text{id}_A \otimes e^\#} & A \otimes A \\ \uparrow e^\# \otimes \text{id}_A & & \uparrow m^\# \\ A \otimes A & \xleftarrow{m^\#} & A \end{array}$$

Antipode identity. Let

$$\mu_A : A \otimes_k A \longrightarrow A$$

be multiplication in A , and let

$$\eta_A : k \longrightarrow A$$

be the unit map. Then

$$\mu_A \circ (i^\# \otimes \text{id}_A) \circ m^\# = \eta_A \circ e^\# = \mu_A \circ (\text{id}_A \otimes i^\#) \circ m^\#.$$

Definition 1.5. A commutative k -algebra A , together with homomorphisms

$$m^\# : A \rightarrow A \otimes_k A, \quad e^\# : A \rightarrow k, \quad i^\# : A \rightarrow A$$

satisfying the preceding coassociativity, counit, and antipode identities, is called a *Hopf algebra*.

Remark (Anti-equivalence of categories). There is an anti-equivalence

$$\left\{ \begin{array}{c} \text{finitely generated commutative} \\ \text{Hopf } k\text{-algebras} \end{array} \right\} \simeq \{ \text{linear algebraic groups over } k \},$$

sending A to $\mathrm{Spm} A$.

Definition 1.6. A *homomorphism of algebraic groups* $\varphi : G \rightarrow G'$ is a morphism of schemes such that

$$\begin{array}{ccc} G \times G & \xrightarrow{(\varphi, \varphi)} & G' \times G' \\ m_G \downarrow & & \downarrow m_{G'} \\ G & \xrightarrow{\varphi} & G' \end{array}$$

commutes.

Definition 1.7. An *algebraic subgroup* H/k of an algebraic group G/k is an algebraic group H/k such that

1. H is a closed k -subscheme of G ;
2. the inclusion $H \hookrightarrow G$ is a homomorphism of algebraic groups.

Remark. In the preceding definition, the word “closed” in (1) is redundant for algebraic subgroup schemes of finite type over a field.

1.1.1 Algebraic groups as functors

Let Alg_k denote the category of finitely generated commutative k -algebras.

Proposition 1.8 (Yoneda). The functor-of-points construction

$$X \longmapsto (R \longmapsto X(R) := \mathrm{Hom}_{\mathrm{Sch}/k}(\mathrm{Spec} R, X))$$

gives a fully faithful functor

$$\{\text{schemes over } k\} \longrightarrow \{\text{functors } \mathbf{Alg}_k \rightarrow \mathbf{Set}\}.$$

The functors in its essential image are called *representable functors*.

Now let G be an algebraic group over k , with multiplication m , unit e , and inverse i . For every $R \in \mathbf{Alg}_k$, these induce

$$m_R : G(R) \times G(R) \longrightarrow G(R), \quad e_R := e \circ p_R \in G(R), \quad i_R : G(R) \longrightarrow G(R),$$

where $p_R : \text{Spec } R \rightarrow \text{Spec } k$ is the structure morphism. The commutative diagrams in the definition of an algebraic group imply that $(G(R), m_R)$ is a group.

Hence an algebraic group gives a functor

$$\tilde{G} : \mathbf{Alg}_k \longrightarrow \mathbf{Grp}, \quad R \longmapsto G(R).$$

Proposition 1.9 (Functorial definition). An algebraic group over k can equivalently be described as a functor

$$\tilde{G} : \mathbf{Alg}_k \longrightarrow \mathbf{Grp}$$

such that the composite

$$\mathbf{Alg}_k \xrightarrow{\tilde{G}} \mathbf{Grp} \longrightarrow \mathbf{Set}$$

is representable.

Exercise 1.10. Check directly that the scheme-theoretic and functorial definitions of an algebraic group coincide.

Remark. For an algebraic group, the unit e and inverse i are uniquely determined by the multiplication m .

Remark. An algebraic group G is commutative if and only if $G(R)$ is a commutative group for every $R \in \mathbf{Alg}_k$.

If k is perfect and G is reduced (equivalently, by the smoothness criterion below, G is smooth), then it is enough to test this on a separable closure k^s :

$$G \text{ is commutative} \iff G(k^s) \text{ is commutative.}$$

Example 1.11 (Basic examples). 1. The trivial group:

$$G = \operatorname{Spm} k.$$

2. The additive group:

$$\mathbb{G}_a : \operatorname{Alg}_k \longrightarrow \mathbf{Grp}, \quad R \longmapsto (R, +),$$

with

$$\mathbb{G}_a = \operatorname{Spm} k[T].$$

Its Hopf algebra structure is

$$m^\# : k[T] \longrightarrow k[T] \otimes_k k[T], \quad T \longmapsto T \otimes 1 + 1 \otimes T,$$

$$e^\# : k[T] \longrightarrow k, \quad T \longmapsto 0,$$

and

$$i^\# : k[T] \longrightarrow k[T], \quad T \longmapsto -T.$$

Thus, if $m^\#(T) = \sum_j f_j \otimes g_j$, then for $x, y \in \mathbb{G}_a$,

$$\sum_j f_j(x)g_j(y) = x + y.$$

3. The multiplicative group:

$$\mathbb{G}_m : \operatorname{Alg}_k \longrightarrow \mathbf{Grp}, \quad R \longmapsto (R^\times, \cdot),$$

with

$$\mathbb{G}_m = \operatorname{Spm} k[T, T^{-1}].$$

Its Hopf algebra structure is

$$m^\# : k[T, T^{-1}] \longrightarrow k[T, T^{-1}] \otimes_k k[T, T^{-1}], \quad T \longmapsto T \otimes T,$$

$$e^\# : k[T, T^{-1}] \longrightarrow k, \quad T \longmapsto 1,$$

and

$$i^\# : k[T, T^{-1}] \longrightarrow k[T, T^{-1}], \quad T \longmapsto T^{-1}.$$

4. Let Γ be a finite group. The associated constant algebraic group is

$$\Gamma = \coprod_{\gamma \in \Gamma} \operatorname{Spec} k = \operatorname{Spm} \left(\bigoplus_{\gamma \in \Gamma} ke_{\gamma} \right),$$

where the e_{γ} are orthogonal idempotents. Writing $\mathcal{O}(\Gamma) = \bigoplus_{\gamma \in \Gamma} ke_{\gamma}$, one has

$$m^{\#} : \mathcal{O}(\Gamma) \longrightarrow \mathcal{O}(\Gamma) \otimes_k \mathcal{O}(\Gamma), \quad e_x \longmapsto \sum_{g \in \Gamma} e_g \otimes e_{g^{-1}x},$$

$$e^{\#} : \mathcal{O}(\Gamma) \longrightarrow k, \quad e_x \longmapsto \begin{cases} 1, & x = e, \\ 0, & x \neq e, \end{cases}$$

and

$$i^{\#} : \mathcal{O}(\Gamma) \longrightarrow \mathcal{O}(\Gamma), \quad e_x \longmapsto e_{x^{-1}}.$$

5. For $n \in \mathbb{N}$,

$$\operatorname{GL}_n \supseteq B_n \supseteq U_n, \quad T_n \subseteq B_n.$$

As functors on Alg_k ,

$$\operatorname{GL}_n(R) = \{(x_{ij}) \in M_n(R) \mid \det(x_{ij}) \in R^{\times}\},$$

$$B_n(R) = \{(x_{ij}) \in \operatorname{GL}_n(R) \mid x_{ij} = 0 \text{ for } i > j\},$$

$$U_n(R) = \{(x_{ij}) \in B_n(R) \mid x_{ii} = 1 \text{ for all } i\},$$

and

$$T_n(R) = \{(x_{ij}) \in B_n(R) \mid x_{ij} = 0 \text{ for } i < j\}.$$

Thus the typical matrix shapes are

$$B_n : \begin{pmatrix} * & * & * \\ 0 & * & * \\ 0 & 0 & * \end{pmatrix}, \quad U_n : \begin{pmatrix} 1 & * & * \\ 0 & 1 & * \\ 0 & 0 & 1 \end{pmatrix}, \quad T_n : \begin{pmatrix} * & 0 & 0 \\ 0 & * & 0 \\ 0 & 0 & * \end{pmatrix}.$$

Here GL_n is a reductive group, B_n is a solvable (Borel, hence parabolic) subgroup of GL_n , U_n is a unipotent group, and T_n is a torus.

1.2 Basic properties

Proposition 1.12. Algebraic groups are separated.

Proof. It is enough to show that the diagonal

$$\Delta_G = \{(x, x) \mid x \in G\} \subseteq G \times G$$

is closed. Consider the morphism

$$G \times G \xrightarrow{(\text{id}_G, i)} G \times G \xrightarrow{m} G, \quad (x, y) \mapsto xy^{-1}.$$

Then

$$\Delta_G = (m \circ (\text{id}_G, i))^{-1}(e),$$

and the identity section $e : \text{Spec } k \rightarrow G$ is a closed point. Hence Δ_G is closed. $\square$

Proposition 1.13. Let G be an algebraic group over k . Then

$$G \text{ is geometrically reduced} \iff G \text{ is smooth over } k.$$

Proof. The implication “smooth $\Rightarrow$ geometrically reduced” is immediate.

For the converse, after extending scalars to an algebraic closure $\bar{k}$, we may assume $k = \bar{k}$. Then G is reduced. The smooth locus of the finite-type k -scheme G is a nonempty open dense subset. Choose a smooth point $g \in G$. For any $h \in G$, left translation by hg^{-1} is an automorphism of G sending g to h . Smoothness is preserved by isomorphisms, so every point of G is smooth. Thus G is smooth. $\square$

Remark. If $\text{char } k = 0$, then every algebraic group over k is smooth.¹

Example 1.14. Assume $\text{char } k = p > 0$. Define

$$\mu_p := \ker \left(\mathbb{G}_m \xrightarrow{[p]} \mathbb{G}_m \right), \quad x \mapsto x^p.$$

¹2026.9.7 stopped at here.

Equivalently, μ_p is the fiber product

$$\begin{array}{ccc} \mu_p & \longrightarrow & \operatorname{Spec} k \\ \downarrow & & \downarrow e \\ \mathbb{G}_m & \xrightarrow{[p]} & \mathbb{G}_m. \end{array}$$

On coordinate rings, the p -power map corresponds to

$$k[T, T^{-1}] \longrightarrow k[T, T^{-1}], \quad T \longmapsto T^p,$$

while the identity section corresponds to

$$k[T, T^{-1}] \longrightarrow k, \quad T \longmapsto 1.$$

Hence

$$\mu_p \simeq \operatorname{Spm} \frac{k[T]}{(T^p - 1)} = \operatorname{Spm} \frac{k[T]}{((T - 1)^p)}.$$

Thus μ_p is nonreduced, and therefore it is not smooth.

Remark (Reduced subscheme). For a scheme X , write X_{red} for its reduced closed subscheme. If $X = \operatorname{Spec} A$, then

$$X_{\text{red}} = \operatorname{Spec}(A/\sqrt{(0)}) \hookrightarrow \operatorname{Spec} A.$$

The universal property used below is that every morphism from a reduced scheme to X factors uniquely through X_{red} .

Recall. A morphism $\varphi : G \rightarrow H$ of algebraic groups is a homomorphism precisely when the multiplication square commutes:

$$\begin{array}{ccc} G \times G & \xrightarrow{m_G} & G \\ \varphi \times \varphi \downarrow & & \downarrow \varphi \\ H \times H & \xrightarrow{m_H} & H. \end{array}$$

Idea. Restrict the multiplication of G to

$$G_{\text{red}} \times G_{\text{red}} \longrightarrow G.$$

To factor this morphism through G_{red} , it is enough that $G_{\text{red}} \times G_{\text{red}}$ be reduced. In particular, this holds when G_{red} is geometrically reduced.

Proposition 1.15. Let G be an algebraic group over k . If G_{red} is geometrically reduced, then G_{red} is an algebraic subgroup of G . In particular, this holds whenever k is perfect.

Proof. Let $j : G_{\text{red}} \hookrightarrow G$ be the canonical closed immersion.

Since $\text{Spec } k$ is reduced, the identity morphism $e : \text{Spec } k \rightarrow G$ factors uniquely through G_{red} :

$$\begin{array}{ccc} \text{Spec } k & \xrightarrow{e} & G \\ & \searrow e_{\text{red}} & \nearrow j \\ & G_{\text{red}} & \end{array}$$

Since G_{red} is reduced, the inverse morphism $\iota \circ j : G_{\text{red}} \rightarrow G$ factors uniquely through G_{red} .

Finally, because G_{red} is geometrically reduced, $G_{\text{red}} \times_k G_{\text{red}}$ is reduced. Hence

$$m_G \circ (j \times j) : G_{\text{red}} \times_k G_{\text{red}} \longrightarrow G$$

factors uniquely through G_{red} :

$$\begin{array}{ccc} G_{\text{red}} \times G_{\text{red}} & \xrightarrow{m_{\text{red}}} & G_{\text{red}} \\ j \times j \downarrow & & \downarrow j \\ G \times G & \xrightarrow{m_G} & G. \end{array}$$

Thus the identity, inverse, and multiplication of G restrict to G_{red} , so G_{red} is an algebraic subgroup of G .

If k is perfect, every reduced finite-type k -scheme is geometrically reduced by [GW20, Proposition 5.49], so the hypothesis is automatic. $\square$

Example 1.16 (Why geometric reducedness matters). Let

$$k = \mathbb{F}_p(t), \quad K = k[x]/(x^p - t), \quad X = \text{Spec } K.$$

Since $t \notin k^p$, the polynomial $x^p - t$ is irreducible over k , so K is a field and X is reduced. If $\beta \in \bar{k}$ satisfies $\beta^p = t$, then

$$X_{\bar{k}} \cong \text{Spec } \frac{\bar{k}[y]}{(y^p - t)} = \text{Spec } \frac{\bar{k}[y]}{(y - \beta)^p},$$

which is not reduced. Thus X is reduced but not geometrically reduced.

If $\alpha \in K$ is the class of x , so that $\alpha^p = t$, then

$$X \times_k X \cong \operatorname{Spec}(K \otimes_k K) \cong \operatorname{Spec} \frac{K[y]}{(y^p - t)} = \operatorname{Spec} \frac{K[y]}{(y - \alpha)^p},$$

which is not reduced.

Definition 1.17 (Kernel). Let $\varphi : G \rightarrow H$ be a homomorphism of algebraic groups. Its kernel is the fiber product

$$\ker(\varphi) = G \times_{H,e} \operatorname{Spec} k,$$

i.e.

$$\begin{array}{ccc} \ker(\varphi) & \longrightarrow & \operatorname{Spec} k \\ \downarrow & & \downarrow e \\ G & \xrightarrow{\varphi} & H. \end{array}$$

Definition 1.18 (Normal algebraic subgroup). Let G be an algebraic group over k , and let $H \subseteq G$ be an algebraic subgroup. Then H is *normal* if, for every $R \in \operatorname{Alg}_k$,

$$H(R) \trianglelefteq G(R).$$

Remark. Let $\varphi : G \rightarrow H$ be a homomorphism. For every $R \in \operatorname{Alg}_k$,

$$(\ker \varphi)(R) = \ker(G(R) \rightarrow H(R)).$$

Hence $\ker \varphi$ is a normal algebraic subgroup of G .

Remark. If G and H are affine, then $\ker \varphi$ is affine. Writing

$$I_H = \ker(e^\# : \mathcal{O}(H) \rightarrow k), \quad \varphi^\# : \mathcal{O}(H) \rightarrow \mathcal{O}(G),$$

one has

$$\mathcal{O}(\ker \varphi) \cong \mathcal{O}(G) \otimes_{\mathcal{O}(H)} k \cong \frac{\mathcal{O}(G)}{\mathcal{O}(G) \varphi^\#(I_H)}.$$

Definition 1.19 (Identity component). Let G be an algebraic group over k . The connected component of G containing the identity e is called the *identity component* of G , and is denoted by G° .

Proposition 1.20 (Largest finite étale subalgebra). Let X be a finite-type k -scheme. Then there exists a largest finite étale k -subalgebra

$$\pi^0(X) \subseteq \mathcal{O}(X).$$

Proof. A finite étale k -algebra has the form

$$\prod_{i=1}^r L_i,$$

with each L_i/k a finite separable field extension. A quotient of a finite étale k -algebra is again finite étale.

If $A_1, A_2 \subseteq \mathcal{O}(X)$ are finite étale k -subalgebras, then their compositum inside $\mathcal{O}(X)$ is the image of

$$A_1 \otimes_k A_2 \longrightarrow \mathcal{O}(X),$$

hence is again finite étale. It therefore suffices to bound $\dim_k A$ for all finite étale k -subalgebras $A \subseteq \mathcal{O}(X)$.

Let k^s be a separable closure of k , and put $n = \dim_k A$. Then

$$A \otimes_k k^s \cong (k^s)^n.$$

Thus

$$1 = e_1 + \cdots + e_n,$$

where the e_i are pairwise orthogonal idempotents. Viewed as functions on X_{k^s} , these idempotents determine n pairwise disjoint nonempty open-and-closed subsets. Consequently,

$$n \leq \#\{\text{connected components of } X_{k^s}\},$$

which is finite because X is of finite type over k .

Choose a finite étale subalgebra of maximal possible k -dimension. Taking its compositum with any other finite étale subalgebra cannot increase the dimension, so it already contains every other one. This gives the largest such subalgebra $\pi^0(X)$. $\square$

Definition 1.21 (The scheme of connected components). Define

$$\pi_0(X) = \text{Spec } \pi^0(X).$$

The inclusion

$$\pi^0(X) \hookrightarrow \mathcal{O}(X)$$

induces a canonical morphism

$$\pi_X : X \longrightarrow \pi_0(X).$$

The scheme $\pi_0(X)$ is finite étale over k .

Proposition 1.22 (Universal property). The morphism $\pi_X : X \rightarrow \pi_0(X)$ is universal for morphisms from X to finite étale k -schemes. Namely, for every morphism $f : X \rightarrow Y$ with Y finite étale over k , there is a unique morphism $\bar{f} : \pi_0(X) \rightarrow Y$ such that

$$f = \bar{f} \circ \pi_X.$$

Equivalently,

$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ & \searrow \pi_X & \nearrow \bar{f} \\ & \pi_0(X) & \end{array}$$

commutes.

Proof. The map f induces

$$f^\# : \mathcal{O}(Y) \longrightarrow \mathcal{O}(X).$$

Its image is a quotient of a finite étale k -algebra, hence is finite étale and therefore lies in $\pi^0(X)$. This gives the factorization, and maximality of $\pi^0(X)$ gives uniqueness. $\square$

Proposition 1.23 (Basic properties of π_0). Let X, Y be finite-type k -schemes.

1. For every field extension k'/k ,

$$\pi_0(X_{k'}) \cong \pi_0(X)_{k'}.$$

- 2.

$$\pi_0(X \times_k Y) \cong \pi_0(X) \times_k \pi_0(Y).$$

3. The fibers of $\pi_X : X \rightarrow \pi_0(X)$ are the connected components of X .

4. For every $x \in \pi_0(X)$, the fiber X_x is geometrically connected over $\kappa(x)$ (not

necessarily over k).

5. The morphism

$$X \longrightarrow \pi_0(X)$$

is faithfully flat.

Remark. See [Mil17, Propositions 1.30, 1.31, 1.33].

Corollary 1.24. Let X be a connected finite-type k -scheme such that $X(k) \neq \emptyset$. Then X is geometrically connected. Moreover, for every connected finite-type k -scheme Y , the product $X \times_k Y$ is connected.

Proof. Since X is connected,

$$\pi_0(X) = \operatorname{Spec} L$$

for some finite separable extension L/k . A k -rational point of X gives a composite

$$\operatorname{Spec} k \longrightarrow X \longrightarrow \operatorname{Spec} L,$$

hence a k -algebra map $L \rightarrow k$. Therefore $L = k$. By the geometric-connectedness property of the fibers, X is geometrically connected over k .

For connected Y ,

$$\pi_0(X \times_k Y) \cong \pi_0(X) \times_k \pi_0(Y) = \operatorname{Spec} k \times_k \pi_0(Y) \cong \pi_0(Y),$$

which is single point. Hence $X \times_k Y$ is connected. □

Remark (The hypothesis $X(k) \neq \emptyset$ is essential). Let

$$X = \operatorname{Spec} L,$$

where L/k is a finite Galois extension with $L \neq k$. Then X is connected, but

$$X_{\bar{k}} = \operatorname{Spec}(L \otimes_k \bar{k}) \cong \coprod_{[L:k]} \operatorname{Spec} \bar{k},$$

so X is not geometrically connected. Also,

$$X \times_k X = \operatorname{Spec}(L \otimes_k L) \cong \coprod_{[L:k]} \operatorname{Spec} L,$$

so $X \times_k X$ is not connected.

Proposition 1.25. Let G be an algebraic group over k . Then:

1. $\pi_0(G)$ is a finite étale algebraic group over k .
2. The canonical morphism

$$\pi : G \longrightarrow \pi_0(G)$$

is a homomorphism of algebraic groups, and it is universal among homomorphisms from G to finite étale algebraic groups. Thus, if H is finite étale and $f : G \rightarrow H$ is a homomorphism, there is a unique homomorphism $\bar{f} : \pi_0(G) \rightarrow H$ such that

$$\begin{array}{ccc} G & \xrightarrow{f} & H \\ & \searrow \pi & \nearrow \bar{f} \\ & \pi_0(G) & \end{array}$$

commutes.

- 3.

$$\ker(G \longrightarrow \pi_0(G)) = G^\circ.$$

In particular, G° is an algebraic subgroup of G .

4. For every field extension k'/k ,

$$\pi_0(G_{k'}) \cong \pi_0(G)_{k'}, \quad (G_{k'})^\circ \cong (G^\circ)_{k'}.$$

In particular, G° is geometrically connected.

5. The fibers of $G \rightarrow \pi_0(G)$ are the connected components of G .
6. For algebraic groups G, G' ,

$$\pi_0(G \times_k G') \cong \pi_0(G) \times_k \pi_0(G'), \quad (G \times_k G')^\circ \cong G^\circ \times_k (G')^\circ.$$

Proof. For the multiplication, consider the composite

$$G \times G \xrightarrow{m_G} G \xrightarrow{\pi} \pi_0(G).$$

By the universal property of π_0 , it factors uniquely through

$$\pi_0(G \times G) \cong \pi_0(G) \times \pi_0(G),$$

giving a multiplication

$$m_0 : \pi_0(G) \times \pi_0(G) \longrightarrow \pi_0(G)$$

such that

$$\begin{array}{ccc} G \times G & \xrightarrow{m_G} & G \\ \pi \times \pi \downarrow & & \downarrow \pi \\ \pi_0(G) \times \pi_0(G) & \xrightarrow{m_0} & \pi_0(G) \end{array}$$

commutes. The identity and inverse descend in the same way:

$$\begin{array}{ccc} \mathrm{Spec} k \xrightarrow{e} G & \xrightarrow{\pi} & \pi_0(G), \\ & \pi \downarrow & \downarrow \pi \\ & \pi_0(G) \xrightarrow{\iota_0} & \pi_0(G). \end{array}$$

These maps satisfy the group axioms, and π is a homomorphism. The universal property among homomorphisms follows from the universal property of π_0 .

For (iii), the kernel is the fiber of π over the identity of $\pi_0(G)$; by the fiber description above, this is exactly the connected component G° .

For (iv), using the identity sections of the component groups,

$$\begin{aligned} (G_{k'})^\circ &= G_{k'} \times_{\pi_0(G_{k'})} \mathrm{Spec} k' \\ &\cong (G \times_{\pi_0(G)} \mathrm{Spec} k) \times_k k' \\ &= (\ker(G \rightarrow \pi_0(G)))_{k'} \\ &= (G^\circ)_{k'}. \end{aligned}$$

The remaining assertions follow from the basic properties of π_0 . □

Definition 1.26 (Component group). The finite étale algebraic group $\pi_0(G)$ is called the *component group*, or the *group of connected components*, of G .

Lemma 1.27. Let G be an algebraic group over k .

1. G is connected if and only if

$$\pi_0(G) = \mathrm{Spec} k.$$

2. If $f : H \rightarrow G$ is a homomorphism and H is connected, then f factors through

G° :

$$\begin{array}{ccc} H & \xrightarrow{f} & G \\ & \searrow \text{dashed} & \nearrow \\ & G^\circ & \end{array}$$

3. The identity component G° is geometrically connected, but the other connected components of G need not be geometrically connected over k .

Proof. For (1), a connected algebraic group has the k -rational identity point, so the preceding corollary gives geometric connectedness and forces its finite étale component scheme to be $\text{Spec } k$. The converse is immediate from the description of the fibers of $G \rightarrow \pi_0(G)$.

For (2), the composite

$$H \longrightarrow G \longrightarrow \pi_0(G)$$

has connected image and contains the identity, so it is the identity morphism into the identity component of the finite étale scheme $\pi_0(G)$, namely the single point $\text{Spec } k$. Hence $H \rightarrow G$ factors through the kernel G° .

Part (3) follows from base change for G° ; the following example shows the second assertion. $\square$

Example 1.28. Let $k = \mathbb{Q}$, and consider

$$\mu_3(R) = \{x \in R^\times : x^3 = 1\}, \quad R \in \text{Alg}_{\mathbb{Q}}.$$

Then

$$\mu_3 = \text{Spec } \frac{\mathbb{Q}[T]}{(T^3 - 1)} \cong \text{Spec } \mathbb{Q} \coprod \text{Spec } \frac{\mathbb{Q}[T]}{(T^2 + T + 1)}.$$

The first component contains the identity. The second component has no $\mathbb{Q}$ -point and is not geometrically connected: after base change to $\overline{\mathbb{Q}}$, it splits into two points. Thus connected components other than G° need not be geometrically connected over the ground field.

Definition 1.29 (Exact sequence of algebraic groups). A sequence

$$e \longrightarrow N \xrightarrow{i} G \xrightarrow{q} Q \longrightarrow e$$

is called exact if:

1. i identifies N isomorphically with $\ker(q : G \rightarrow Q)$;

2. q is faithfully flat.

Example 1.30. The sequence

$$e \longrightarrow G^\circ \longrightarrow G \longrightarrow \pi_0(G) \longrightarrow e$$

is exact.

Proposition 1.31. Every connected algebraic group is irreducible. In fact, it is geometrically irreducible.

Proof. It is enough to prove that G is geometrically irreducible. Since a connected algebraic group is geometrically connected, after base change we may assume $k = \bar{k}$.

The reduced closed subscheme $G_{\text{red}} \subseteq G$ has the same underlying topological space:

$$|G_{\text{red}}| = |G|.$$

Since k is perfect, G_{red} is an algebraic subgroup of G , and a reduced algebraic group over a perfect field is smooth. Thus, for the topological question of irreducibility, we may replace G by the smooth reduced group G_{red} .

Suppose G had more than one irreducible component. Because G is connected, two irreducible components must meet. Choose a point x in such an intersection. Then the local ring $\mathcal{O}_{G,x}$ has at least two distinct minimal prime ideals. On the other hand, smoothness implies that $\mathcal{O}_{G,x}$ is a regular local ring, hence an integral domain, so it has only one minimal prime. This is a contradiction. $\square$

Proposition 1.32. Let $\varphi : G \rightarrow H$ be a homomorphism of algebraic groups. Then the image $\varphi(|G|) \subseteq |H|$ is closed.

Proof. Step 1. It is enough to reduce to the case $k = \bar{k}$.

First, for any finite-type k -scheme X , the projection

$$p_X : |X_{\bar{k}}| \longrightarrow |X|$$

is a quotient map of topological spaces.

Second, if

$$p_H : |H_{\bar{k}}| \longrightarrow |H|$$

is the projection, then

$$\varphi_{\bar{k}}(|G_{\bar{k}}|) = p_H^{-1}(\varphi(|G|)).$$

Therefore $\varphi(|G|)$ is closed in $|H|$ if and only if the image after base change to $\bar{k}$ is closed in $|H_{\bar{k}}|$. Hence the proposition reduces to the algebraically closed case.²

Step 2. It is enough to work after passing to an algebraic closure, so assume $k = \bar{k}$. Then $|G| = G(k)$ on closed points. Put

$$N := \varphi(G(k)) \subseteq H(k),$$

and let $\overline{N}$ denote its closure in $|H|$. We show first that $\overline{N}$ is a subgroup of $H(k)$.

Let

$$i : H \xrightarrow{\sim} H, \quad x \mapsto x^{-1}.$$

Since i is a homeomorphism and $N^{-1} = N$,

$$(\overline{N})^{-1} = i(\overline{N}) = \overline{i(N)} = \overline{N}.$$

For $x \in N$, left translation

$$\ell_x : H \xrightarrow{\sim} H, \quad y \mapsto xy$$

satisfies

$$\ell_x(\overline{N}) = \overline{\ell_x(N)} = \overline{xN} = \overline{N}.$$

Hence $N\overline{N} \subseteq \overline{N}$. Now let $x \in \overline{N}$. Right translation

$$r_x : H \xrightarrow{\sim} H, \quad y \mapsto yx$$

gives

$$r_x(\overline{N}) = \overline{r_x(N)} = \overline{Nx} \subseteq \overline{N},$$

because $Nx \subseteq N\overline{N} \subseteq \overline{N}$. Thus

$$\overline{N}\overline{N} \subseteq \overline{N},$$

so $\overline{N}$ is a subgroup of $H(k)$.

Remark. This is the same idea as in the standard topological-group argument.

By Chevalley's theorem below, there exists a nonempty open dense subset

$$U \subseteq \overline{N}$$

²2026.9.10 stopped at here.

with $U \subseteq N$. For any $x \in \overline{N}$, both U and $x^{-1}U$ are open dense subsets of $\overline{N}$. Hence

$$U \cap x^{-1}U \neq \emptyset.$$

Choose u in this intersection. Then $u \in U$ and $xu \in U$, so

$$x = (xu)u^{-1} \in UU^{-1} \subseteq N.$$

Therefore $\overline{N} \subseteq N$, hence $N = \overline{N}$ is closed. □

Theorem 1.33 (Chevalley). Let $f : Y \rightarrow X$ be a finite-type morphism of noetherian schemes. Then $f(|Y|)$ is constructible in $|X|$. In particular, $f(|Y|)$ contains an open dense subset of its closure.

Corollary 1.34. If $\varphi : G \rightarrow H$ is a dominant homomorphism of algebraic groups, then φ is surjective on underlying topological spaces.

Proof. By the proposition, $\varphi(|G|)$ is closed in $|H|$. Dominance says that it is dense, hence it equals $|H|$. □

Proposition 1.35. Let $\varphi : G \rightarrow H$ be a surjective homomorphism of algebraic groups. If H is reduced, then φ is faithfully flat.

Proof. By generic flatness there is an open dense subset $U \subseteq H$ such that

$$\varphi^{-1}(U) \longrightarrow U$$

is flat. After extending scalars to an algebraic closure $\overline{k}$, the corresponding morphism over $U_{\overline{k}}$ is still flat.

For $g \in G(\overline{k})$, left translations give a commutative diagram

$$\begin{array}{ccc} G_{\overline{k}} & \xrightarrow[\sim]{L_g} & G_{\overline{k}} \\ \varphi_{\overline{k}} \downarrow & & \downarrow \varphi_{\overline{k}} \\ H_{\overline{k}} & \xrightarrow[\sim]{L_{\varphi(g)}} & H_{\overline{k}}. \end{array}$$

Thus $\varphi_{\overline{k}}$ is flat over every translate $\varphi(g)U_{\overline{k}}$. Surjectivity gives

$$H_{\overline{k}} = \bigcup_{g \in G(\overline{k})} \varphi(g)U_{\overline{k}},$$

so $\varphi_{\bar{k}}$ is flat everywhere. Flatness descends along the faithfully flat extension $\bar{k}/k$, hence φ is flat. Since φ is surjective, it is faithfully flat. $\square$

1.3 The image of a homomorphism

Motivation. For a homomorphism of abstract groups $G \xrightarrow{\varphi} H$, one factors φ through its image $\varphi(G)$. The analogous statement for algebraic groups is the following.

Theorem 1.36. Let $\varphi : G \rightarrow H$ be a homomorphism of algebraic groups, and suppose that G is smooth. Then φ factors as

$$\begin{array}{ccc} G & \xrightarrow{\varphi} & H \\ & \searrow q & \nearrow i \\ & I & \end{array}$$

where q is faithfully flat and i is a closed immersion.

Remark. The role of the smoothness hypothesis in this proof is to ensure that $G_{\bar{k}}$ is reduced; this forces the reduced image I to be geometrically reduced, which is what allows the group operations to restrict to I .

Proof. Let

$$I := \varphi(G)_{\text{red}}$$

with its reduced induced closed-subscheme structure in H . Since G is reduced, φ factors through I ; write the factorization as $G \xrightarrow{q} I \xrightarrow{i} H$. The map q is surjective, and I is reduced, so Proposition 1.35 shows that q is faithfully flat.

It remains to see that I is an algebraic subgroup of H . It is enough to prove that I is geometrically reduced. After base change to $\bar{k}$,

$$q_{\bar{k}} : G_{\bar{k}} \longrightarrow I_{\bar{k}}$$

is faithfully flat. Let $x \in |I_{\bar{k}}|$ and choose $g \in |G_{\bar{k}}|$ mapping to x . Then

$$\mathcal{O}_{I_{\bar{k}},x} \longrightarrow \mathcal{O}_{G_{\bar{k}},g}$$

is a faithfully flat local homomorphism, hence injective. Since G is smooth, $G_{\bar{k}}$ is reduced, so $\mathcal{O}_{G_{\bar{k}},g}$ is reduced. Therefore $\mathcal{O}_{I_{\bar{k}},x}$ is reduced. As x was arbitrary, $I_{\bar{k}}$ is reduced, hence I is geometrically reduced.

Now the multiplication and inverse maps of H carry the underlying sets $|I| \times |I|$ and $|I|$ into $|I|$. Since I and $I \times I$ are geometrically reduced, these morphisms factor through the

reduced closed subscheme I . Thus I inherits an algebraic-group structure and i is a closed immersion of algebraic groups. $\square$

Definition 1.37. The algebraic subgroup I above is called the *image* of φ .

Remark. For linear algebraic groups, an image factorization of this form exists without assuming that G is smooth.

1.4 Group actions

Definition 1.38. An action of an algebraic group G on a k -scheme X is a morphism

$$\varphi : G \times_k X \longrightarrow X$$

such that the following two conditions hold.

1. If $e : \operatorname{Spec} k \rightarrow G$ is the identity section, then

$$\begin{array}{ccc} X = \operatorname{Spec} k \times X & \xrightarrow{e \times \operatorname{id}_X} & G \times X \\ & \searrow \operatorname{id}_X & \downarrow \varphi \\ & & X \end{array}$$

commutes.

2. If $m : G \times G \rightarrow G$ is multiplication, then

$$\begin{array}{ccc} G \times G \times X & \xrightarrow{m \times \operatorname{id}_X} & G \times X \\ \operatorname{id}_G \times \varphi \downarrow & & \downarrow \varphi \\ G \times X & \xrightarrow{\varphi} & X \end{array}$$

commutes.

Remark. By Yoneda's lemma, an action of G on X is equivalently a natural transformation

$$\tilde{G} \times \tilde{X} \longrightarrow \tilde{X}$$

such that $G(R)$ acts on $X(R)$ for every k -algebra R .

Example 1.39. 1. The trivial action:

$$G \times X \xrightarrow{\text{pr}_X} X, \quad g \cdot x = x.$$

2. For $X = G$, left multiplication:

$$G \times G \xrightarrow{m} G, \quad g \cdot x = gx.$$

3. For $X = G$, right multiplication written as a left action:

$$G \times G \longrightarrow G, \quad g \cdot x = xg^{-1}.$$

4. For $X = G$, conjugation:

$$G \times G \longrightarrow G, \quad g \cdot x = gxg^{-1}.$$

Definition 1.40. A linear representation of G is a homomorphism of algebraic groups

$$\rho : G \longrightarrow \text{GL}_n$$

for some $n \geq 1$.

Remark. If $\rho : G \rightarrow \text{GL}_n$ is a representation, then ρ gives an action of G on $\mathbb{A}^n$. On R -points,

$$G(R) \times \mathbb{A}^n(R) \longrightarrow \mathbb{A}^n(R), \quad \left(g, \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} \right) \longmapsto \rho(g) \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}$$

for every k -algebra R .

Lemma 1.41. Let $\varphi : G \times X \rightarrow X$ be an action. Then φ is faithfully flat. Moreover, φ is smooth if G is smooth over k , and φ is finite if G is finite over k .

Proof. The map

$$\alpha : G \times X \xrightarrow{\sim} G \times X, \quad (g, x) \longmapsto (g, gx)$$

is an isomorphism, with inverse $(g, y) \mapsto (g, g^{-1}y)$. It satisfies

$$\varphi = \text{pr}_X \circ \alpha.$$

Thus every property listed in the statement for φ is the corresponding property of the projection $\text{pr}_X : G \times X \rightarrow X$, which is the base change of $G \rightarrow \text{Spec } k$. $\square$

Definition 1.42. Let $\varphi : G \times X \rightarrow X$ be an action and let $x \in X(k)$. The *orbit map* is

$$\varphi_x : G \simeq G \times \text{Spec } k \xrightarrow{\text{id}_G \times x} G \times X \xrightarrow{\varphi} X, \quad g \mapsto gx.$$

The orbit of x is the image of $|\varphi_x|$ and is denoted by $G \cdot x$.

Lemma 1.43. The orbit $G \cdot x$ is a locally closed subset of $|X|$.

Proof. As in the earlier reduction, it is enough to work after passing to an algebraic closure, so assume $k = \bar{k}$; the same reduction uses that $|X_{\bar{k}}| \rightarrow |X|$ is a quotient map. By Chevalley's theorem applied to $\varphi_x : G \rightarrow X$, there is a nonempty subset

$$U \subseteq G \cdot x$$

which is open and dense in $\overline{G \cdot x}$. Since U is nonempty and contained in the orbit,

$$G \cdot x = G \cdot U = \bigcup_{g \in G(k)} gU.$$

For every $g \in G(k)$, translation by g preserves $\overline{G \cdot x}$, and gU is open in $\overline{G \cdot x}$. Hence $G \cdot x$ is open in its closure, i.e. it is locally closed. $\square$

Remark. It is possible that $X(k) = \emptyset$. For example, let

$$k = \mathbb{Q}, \quad G = (\mathbb{Z}/2\mathbb{Z})_k, \quad X = \text{Spec } \mathbb{Q}(\sqrt{2}).$$

Then G acts on X by Galois conjugation, while $X(\mathbb{Q}) = \emptyset$.

Proposition 1.44. Assume $k = \bar{k}$ and that G acts on X . Every orbit of minimum dimension is closed. In particular, if $X \neq \emptyset$, a closed orbit exists.

Proof. For a subset $X_1 \subseteq X$, write

$$G \cdot X_1 := \{gx \in X : g \in G(k), x \in X_1\}.$$

First note that the closure of an orbit is G -stable, hence is a union of orbits. Indeed, for every $g \in G(k)$, the map

$$\ell_g : X \xrightarrow{\sim} X, \quad x \mapsto gx$$

is an isomorphism, and therefore

$$\ell_g(\overline{G \cdot x}) = \overline{\ell_g(G \cdot x)} = \overline{g(G \cdot x)} = \overline{G \cdot x}.$$

Thus $G \cdot \overline{G \cdot x} = \overline{G \cdot x}$.

Now let

$$y \in \overline{G \cdot x} \setminus G \cdot x.$$

Then

$$G \cdot y \subseteq \overline{G \cdot x} \setminus G \cdot x.$$

By Lemma 1.43, $G \cdot x$ is open and dense in $\overline{G \cdot x}$. Hence its boundary contains no irreducible component of $\overline{G \cdot x}$, and therefore has strictly smaller dimension. Since $G \cdot y$ is contained in the boundary,

$$\dim(G \cdot y) < \dim(G \cdot x).$$

Therefore an orbit of minimum dimension has empty boundary and is closed. $\square$

1.5 Structure of general algebraic groups

Remark. Reference: [Mil17, Chapter 8].

Recall. For an algebraic group G over k , there is an exact sequence

$$e \longrightarrow G^\circ \longrightarrow G \longrightarrow \pi_0(G) \longrightarrow e,$$

where G° is the connected identity component and $\pi_0(G)$ is the finite étale component group.

Theorem 1.45 (Barsotti–Chevalley). Let G be a connected algebraic group over a field k . Then there exists an exact sequence

$$e \longrightarrow N \longrightarrow G \longrightarrow Q \longrightarrow e,$$

where N is a connected affine normal algebraic subgroup and Q is an abelian variety. If, in addition, G is smooth and k is perfect, then N is the unique connected affine normal subgroup variety with abelian-variety quotient.

Remark. This is the pair of statements referred to [Mil17, Theorem 8.27 and 8.28].

Recall. An abelian variety is an algebraic group that is connected, smooth, and complete. In particular, every abelian variety is commutative.

Theorem 1.46. Every algebraic group G over k has a greatest (maximal) affine algebraic quotient

$$G \longrightarrow G_{\text{aff}},$$

universal among homomorphisms from G to affine algebraic groups.³

Remark. More generally, every algebraic group G has a maximal affine quotient. Write

$$G_{\text{aff}} := \text{Spm } \mathcal{O}(G).$$

The first question is whether $\mathcal{O}(G)$ carries the expected Hopf algebra structure. The multiplication, identity, and inverse of G induce

$$m^\# : \mathcal{O}(G) \longrightarrow \mathcal{O}(G \times G) \cong \mathcal{O}(G) \otimes_k \mathcal{O}(G), \quad e^\# : \mathcal{O}(G) \longrightarrow k, \quad i^\# : \mathcal{O}(G) \longrightarrow \mathcal{O}(G),$$

and these maps make $\mathcal{O}(G)$ a Hopf k -algebra.

Theorem 1.47. Let G be an algebraic group over k . Then:

(i) $\mathcal{O}(G)$ is a finitely generated k -algebra. In particular, $G_{\text{aff}} = \text{Spm } \mathcal{O}(G)$ is a linear algebraic group.

(ii) The natural morphism

$$\phi : G \longrightarrow G_{\text{aff}}$$

is universal among homomorphisms from G to linear algebraic groups: if $\phi' : G \rightarrow H$ is a homomorphism and H is a linear algebraic group, then there is a unique homomorphism $\psi : G_{\text{aff}} \rightarrow H$ such that $\phi' = \psi \circ \phi$.

(iii) The morphism ϕ is faithfully flat.

(iv) The kernel $\ker(\phi)$ is anti-affine, i.e.

$$\mathcal{O}(\ker \phi) = k.$$

In particular, $\ker(\phi)$ is commutative.

Remark. Reference. [Mil17, Proposition 8.36, Corollary 8.13] and [Dem70, Exp. VI, Theorem 12.2].

³2026.9.14 stopped at here.

Consequence. There is an exact sequence

$$e \longrightarrow G_{\text{ant}} \longrightarrow G \longrightarrow G_{\text{aff}} \longrightarrow e,$$

where G_{ant} is anti-affine (hence commutative) and G_{aff} is a linear algebraic group.

Consequence. For every linear representation

$$r : G \longrightarrow \text{GL}_n,$$

there is a unique factorization

$$\begin{array}{ccc} G & \xrightarrow{\quad r \quad} & \text{GL}_n \\ & \searrow & \nearrow \text{dashed} \\ & G_{\text{aff}} & \end{array}$$

Thus the study of linear representations of G is equivalent to the study of linear representations of G_{aff} . In what follows, we focus on linear algebraic groups.

2 Linear representations of linear algebraic groups

2.1 Hopf algebras

Recall. Let $G = \text{Spm } A$ be a linear algebraic group over k . Then A has a Hopf algebra structure

$$m^\# : A \longrightarrow A \otimes_k A, \quad e^\# : A \longrightarrow k, \quad i^\# : A \longrightarrow A.$$

Geometrically, if

$$m^\#(f) = \sum_i g_i \otimes h_i,$$

then for every $R \in \text{Alg}_k$ and $x, y \in G(R)$,

$$f(xy) = \sum_i g_i(x)h_i(y) \in R.$$

Moreover,

$$e^\#(f) = f(e), \quad i^\#(f)(x) = f(x^{-1}).$$

Definition 2.1. Let A and B be Hopf k -algebras. A *homomorphism of Hopf algebras* $f : A \rightarrow B$ is a homomorphism of k -algebras such that

$$\begin{array}{ccc} A & \xrightarrow{f} & B \\ m_A^\# \downarrow & & \downarrow m_B^\# \\ A \otimes_k A & \xrightarrow{f \otimes f} & B \otimes_k B \end{array}$$

commutes.

Remark. For such an f , the counit and antipode are automatically compatible as well:

$$e_B^\# \circ f = e_A^\#, \quad i_B^\# \circ f = f \circ i_A^\#.$$

Remark. Homomorphisms of linear algebraic groups

$$\text{Spm } B \longrightarrow \text{Spm } A$$

correspond contravariantly to homomorphisms of Hopf algebras

$$A \longrightarrow B.$$

Definition 2.2. Let B be a Hopf algebra over k . A k -subalgebra $A \subseteq B$ is called a *Hopf subalgebra* if

$$m_B^\#(A) \subseteq A \otimes_k A, \quad i_B^\#(A) \subseteq A.$$

Proposition 2.3. Let $A \subseteq B$ be a Hopf subalgebra. Then A is a Hopf algebra with

$$m_A^\# = m_B^\#|_A, \quad e_A^\# = e_B^\#|_A, \quad i_A^\# = i_B^\#|_A.$$

Moreover, if $f : A \rightarrow B$ is a homomorphism of Hopf algebras, then $f(A)$ is a Hopf subalgebra of B .

Definition 2.4. Let A be a Hopf k -algebra. A *Hopf ideal* is an ideal $\mathfrak{a} \subseteq A$ such that

$$m^\#(\mathfrak{a}) \subseteq \mathfrak{a} \otimes_k A + A \otimes_k \mathfrak{a}, \quad i^\#(\mathfrak{a}) \subseteq \mathfrak{a}, \quad e^\#(\mathfrak{a}) = 0.$$

Equivalently, the composite

$$\mathfrak{a} \hookrightarrow A \xrightarrow{m^\#} A \otimes_k A \longrightarrow (A/\mathfrak{a}) \otimes_k (A/\mathfrak{a})$$

is zero, together with the antipode and counit conditions above.

Lemma 2.5. If $f : A \rightarrow B$ is a homomorphism of Hopf algebras, then $\ker f$ is a Hopf ideal of A .

Proof. Let $\mathfrak{a} = \ker f$. If $a \in \mathfrak{a}$, then

$$(f \otimes f)m_A^\#(a) = m_B^\#(f(a)) = 0.$$

Over the field k ,

$$\ker(f \otimes f) = \mathfrak{a} \otimes_k A + A \otimes_k \mathfrak{a},$$

so

$$m_A^\#(\mathfrak{a}) \subseteq \mathfrak{a} \otimes_k A + A \otimes_k \mathfrak{a}.$$

The compatibility of f with the antipode and counit gives $i_A^\#(\mathfrak{a}) \subseteq \mathfrak{a}$ and $e_A^\#(\mathfrak{a}) = 0$. $\square$

Proposition 2.6. Let $\mathfrak{a}$ be a Hopf ideal of a Hopf k -algebra A .

(i) The quotient $A/\mathfrak{a}$ has a Hopf algebra structure for which $A \rightarrow A/\mathfrak{a}$ is a homo-

morphism of Hopf algebras.

- (ii) If $f : A \rightarrow B$ is a homomorphism of Hopf k -algebras, then the canonical map induces an isomorphism of Hopf algebras

$$A/\ker f \xrightarrow{\sim} f(A).$$

In particular, f factors as

$$A \twoheadrightarrow A/\ker f \xrightarrow{\sim} f(A) \hookrightarrow B.$$

Fact. Let $A \subseteq B$ be finitely generated Hopf k -algebras. Then B is faithfully flat over A . See [Mil17, Theorem 3.31].

Corollary 2.7. Every homomorphism

$$\varphi : G \longrightarrow H$$

of linear algebraic groups factors as

$$G \xrightarrow{q} I \xrightarrow{\iota} H,$$

where q is faithfully flat and ι is a closed immersion.

Corollary 2.8. Let $\varphi : G \rightarrow H$ be a homomorphism of linear algebraic groups. If

$$\ker \varphi = \operatorname{Spm} k,$$

then φ is a closed immersion.

Proof. By the preceding corollary, it is enough to treat the faithfully flat factor q . Thus we may assume that φ itself is faithfully flat, and it remains to show that φ is an isomorphism.

Since $\ker \varphi = \operatorname{Spm} k$, for every $R \in \operatorname{Alg}_k$ the map

$$\varphi(R) : G(R) \longrightarrow H(R)$$

is injective. We show that it is also surjective.

Take $y \in H(R)$. Since φ is faithfully flat, there is a faithfully flat R -algebra R' and an

element $x \in G(R')$ whose image in $H(R')$ is the pullback of y . The descent diagram is

$$\begin{array}{ccccc} G(R) & \longrightarrow & G(R') & \rightrightarrows & G(R' \otimes_R R') \\ \varphi(R) \downarrow & & \varphi(R') \downarrow & & \downarrow \varphi(R' \otimes_R R') \\ H(R) & \longrightarrow & H(R') & \rightrightarrows & H(R' \otimes_R R'). \end{array}$$

The two pullbacks of x to $G(R' \otimes_R R')$ have the same image under $\varphi(R' \otimes_R R')$, because both come from y . Since $\varphi(R' \otimes_R R')$ is injective, the two pullbacks of x agree. Faithfully flat descent for the affine scheme G therefore gives a unique element of $G(R)$ descending to x , and its image under $\varphi(R)$ is y .

Thus $\varphi(R)$ is bijective for every $R \in \text{Alg}_k$. By Yoneda, φ is an isomorphism. $\square$

Remark. A faithfully flat homomorphism of linear algebraic groups $\varphi : G \rightarrow H$ need not be surjective on R -points for every $R \in \text{Alg}_k$ when $\ker \varphi \neq \text{Spm } k$.

Proof. Take $k = \mathbb{R}$, $G = H = \mathbb{G}_m$, and

$$\varphi : \mathbb{G}_m \longrightarrow \mathbb{G}_m, \quad x \longmapsto x^2.$$

On coordinate rings,

$$\varphi^\# : \mathbb{R}[T, T^{-1}] \longrightarrow \mathbb{R}[T, T^{-1}], \quad T \longmapsto T^2,$$

and

$$\varphi^\#(\mathbb{R}[T, T^{-1}]) = \mathbb{R}[T^2, T^{-2}].$$

The ring $\mathbb{R}[T, T^{-1}]$ is a free $\mathbb{R}[T^2, T^{-2}]$ -module with basis $\{1, T\}$, so φ is faithfully flat. However,

$$\varphi(\mathbb{R}) : \mathbb{R}^\times \longrightarrow \mathbb{R}^\times, \quad x \longmapsto x^2,$$

is not surjective. $\square$

Let V be a k -vector space. Define the group-valued functor

$$\text{GL}_V : \text{Alg}_k \longrightarrow \text{Grp}, \quad R \longmapsto \text{GL}_V(R) := \text{Aut}_R(V \otimes_k R).$$

If V is finite-dimensional, then after choosing a basis of V ,

$$\text{GL}_V \cong \text{GL}_n, \quad n = \dim_k V,$$

where the displayed isomorphism depends on the choice of basis.

2.2 Linear representations

Definition 2.9. Let G be a linear algebraic group over k and let V be a k -vector space. A *linear representation* of G on V is a homomorphism of algebraic groups

$$r : G \longrightarrow \mathrm{GL}_V.$$

The representation r is *faithful* if

$$\ker r = \mathrm{Spm} k,$$

and it is *trivial* if its image is the trivial group $\mathrm{Spm} k$.

Remark. For a representation $r : G \rightarrow \mathrm{GL}_V$, the following are equivalent:

- r is faithful;
- $r(R) : G(R) \longrightarrow \mathrm{GL}_V(R)$ is injective for every $R \in \mathrm{Alg}_k$;
- r is a closed immersion.

Construction. Let $r : G \rightarrow \mathrm{GL}_V$ be a representation and write $G = \mathrm{Spm} A$. Since

$$G(A) = \mathrm{Hom}_{k\text{-alg}}(A, A),$$

the identity map id_A defines an A -point of G . Hence

$$r(A)(\mathrm{id}_A) \in \mathrm{Aut}_A(V \otimes_k A).$$

Define a k -linear map

$$\rho_r : V \longrightarrow V \otimes_k A, \quad \rho_r(v) := r(A)(\mathrm{id}_A)(v \otimes 1).$$

Remark. Let $R \in \mathrm{Alg}_k$ and $x \in G(R)$, corresponding to a k -algebra map

$$x^\# : A \longrightarrow R.$$

Then naturality gives

$$\begin{array}{ccc} V & \xrightarrow{\rho_r} & V \otimes_k A \\ & \searrow v \mapsto r(R)(x)(v \otimes 1) & \downarrow \mathrm{id}_V \otimes x^\# \\ & & V \otimes_k R. \end{array}$$

Equivalently, the R -linearization of

$$(\mathrm{id}_V \otimes x^\#) \circ \rho_r : V \longrightarrow V \otimes_k R$$

is exactly $r(R)(x) : V \otimes_k R \rightarrow V \otimes_k R$.

2.3 Comodules

Definition 2.10. A *right $\mathcal{O}(G)$ -comodule* is a k -vector space V together with a k -linear map

$$\rho : V \longrightarrow V \otimes_k \mathcal{O}(G)$$

such that

$$(\mathrm{id}_V \otimes e^\#) \circ \rho = \mathrm{id}_V$$

and

$$(\mathrm{id}_V \otimes m^\#) \circ \rho = (\rho \otimes \mathrm{id}_{\mathcal{O}(G)}) \circ \rho.$$

Proposition 2.11. The map ρ_r satisfies

$$(\mathrm{id}_V \otimes e^\#) \circ \rho_r = \mathrm{id}_V$$

and

$$(\rho_r \otimes \mathrm{id}_A) \circ \rho_r = (\mathrm{id}_V \otimes m^\#) \circ \rho_r.$$

Hence (V, ρ_r) is a right A -comodule.

Proof. The first identity is the coordinate-ring form of the identity action $e \cdot v = v$. The second is the coordinate-ring form of associativity of the action,

$$g_1(g_2v) = (g_1g_2)v.$$

Indeed, applying the construction to the universal points and using that r is a group homomorphism gives exactly the two displayed identities.⁴ $\square$

Definition 2.12. Let (V, ρ) be an $\mathcal{O}(G)$ -comodule. An $\mathcal{O}(G)$ -subcomodule of V is a

⁴2026.9.17 stopped at here.

k -subspace $W \subseteq V$ such that

$$\rho(W) \subseteq W \otimes \mathcal{O}(G).$$

Then $(W, \rho|_W)$ is again an $\mathcal{O}(G)$ -comodule.

Lemma 2.13. For a k -vector space V , there is a bijection

$$\left\{ \text{representations } r : G \rightarrow \mathrm{GL}_V \right\} \longleftrightarrow \left\{ \mathcal{O}(G)\text{-comodule structures on } V \right\},$$

sending r to ρ_r .

Proof. It remains to define the inverse map. Let

$$\rho : V \longrightarrow V \otimes \mathcal{O}(G)$$

be a comodule map. For $R \in \mathrm{Alg}_k$ and

$$g \in G(R) = \mathrm{Hom}_{k\text{-alg}}(\mathcal{O}(G), R),$$

consider

$$V \xrightarrow{\rho} V \otimes \mathcal{O}(G) \xrightarrow{\mathrm{id}_V \otimes g} V \otimes R.$$

Its R -linearization is an element of

$$\mathrm{Aut}_R(V \otimes R) = \mathrm{GL}_V(R).$$

This defines $r_\rho(R) : G(R) \rightarrow \mathrm{GL}_V(R)$. The comodule identities give the identity and multiplication laws, and the two constructions are inverse to each other. $\square$

Example 2.14 (The standard representation of GL_n). Let $G = \mathrm{GL}_n$, let $V = k^n$ with basis $\{e_1, \dots, e_n\}$, and let

$$r : \mathrm{GL}_n \xrightarrow{\sim} \mathrm{GL}_V$$

be the standard representation. Write

$$\mathcal{O}(G) = k[T_{ij}, \det(T)^{-1}].$$

If

$$\rho_r(e_i) = \sum_{j=1}^n e_j \otimes f_{ji},$$

then for $R \in \text{Alg}_k$ and $g = (a_{st}) \in G(R)$,

$$ge_i = \sum_{j=1}^n a_{ji} e_j.$$

Hence $f_{ji}(g) = a_{ji}$, so $f_{ji} = T_{ji}$. Therefore

$$\rho_r(e_i) = \sum_{j=1}^n e_j \otimes T_{ji}.$$

Exercise 2.15. Compute the $\mathcal{O}(G)$ -comodule structures on $V^\vee$ and on $V \otimes V$.

Proposition 2.16. Every $\mathcal{O}(G)$ -comodule (V, ρ) is a filtered union of its finite-dimensional subcomodules.

Proof. It is enough to show that every $v \in V$ is contained in a finite-dimensional subcomodule. Let $\{f_i\}_{i \in I}$ be a k -basis of $\mathcal{O}(G)$. Write

$$\rho(v) = \sum_{i \in I_0} v_i \otimes f_i$$

for a finite subset $I_0 \subset I$, and let

$$W = \text{span}_k\{v_i : i \in I_0\}.$$

The counit identity gives

$$v = (\text{id}_V \otimes e^\#)\rho(v) = \sum_{i \in I_0} v_i e^\#(f_i) \in W,$$

and $\dim_k W < \infty$.

It remains to show that $\rho(W) \subseteq W \otimes \mathcal{O}(G)$. For each $i \in I_0$, write

$$m^\#(f_i) = \sum_{j, \ell \in I} a_{ij\ell} f_j \otimes f_\ell,$$

where only finitely many coefficients occur. Coassociativity gives the commutative square

$$\begin{array}{ccc} V & \xrightarrow{\rho} & V \otimes \mathcal{O}(G) \\ \rho \downarrow & & \downarrow \text{id}_V \otimes m^\# \\ V \otimes \mathcal{O}(G) & \xrightarrow{\rho \otimes \text{id}} & V \otimes \mathcal{O}(G) \otimes \mathcal{O}(G). \end{array}$$

Applying the two paths to v , one obtains

$$\sum_i v_i \otimes m^\#(f_i) = \sum_{i,j,\ell} a_{ij\ell} v_i \otimes f_j \otimes f_\ell$$

and

$$\sum_\ell \rho(v_\ell) \otimes f_\ell.$$

Comparing coefficients of the basis elements f_ℓ in the third tensor factor gives

$$\rho(v_\ell) = \sum_{i,j} a_{ij\ell} v_i \otimes f_j \in W \otimes \mathcal{O}(G).$$

Thus W is a finite-dimensional subcomodule containing v . Since the sum of two finite-dimensional subcomodules is again a finite-dimensional subcomodule, these subcomodules form a filtered system. $\square$

Corollary 2.17. Every representation of G is a filtered union of its finite-dimensional subrepresentations.

Proof. This is the preceding proposition under the representation–comodule correspondence. $\square$

2.4 LAG is linear

Proposition 2.18 (Affine right actions and comodule algebras). Let X be an affine k -scheme. A k -algebra homomorphism

$$\rho : \mathcal{O}(X) \longrightarrow \mathcal{O}(X) \otimes \mathcal{O}(G)$$

which makes $\mathcal{O}(X)$ an $\mathcal{O}(G)$ -comodule corresponds contravariantly to a right action

$$a : X \times G \longrightarrow X.$$

Proof. The counit condition is the commutativity of

$$\begin{array}{ccc} \mathcal{O}(X) & \xrightarrow{\rho} & \mathcal{O}(X) \otimes \mathcal{O}(G) \\ & \searrow \text{id} & \downarrow \text{id} \otimes e^\# \\ & & \mathcal{O}(X), \end{array}$$

which is dual to the commutative triangle

$$\begin{array}{ccc} X \times G & \xrightarrow{a} & X \\ \uparrow (\text{id}_X, e) & \nearrow \text{id}_X & \\ X & & \end{array}$$

for the identity axiom of a right action. The coassociativity condition is

$$\begin{array}{ccc} \mathcal{O}(X) & \xrightarrow{\rho} & \mathcal{O}(X) \otimes \mathcal{O}(G) \\ \rho \downarrow & & \downarrow \text{id} \otimes m^\# \\ \mathcal{O}(X) \otimes \mathcal{O}(G) & \xrightarrow{\rho \otimes \text{id}} & \mathcal{O}(X) \otimes \mathcal{O}(G) \otimes \mathcal{O}(G), \end{array}$$

which is dual to

$$\begin{array}{ccc} X \times G \times G & \xrightarrow{a \times \text{id}_G} & X \times G \\ \text{id}_X \times m \downarrow & & \downarrow a \\ X \times G & \xrightarrow{a} & X. \end{array}$$

Thus the two sets of axioms are equivalent. $\square$

Remark. If $r : G \rightarrow \text{GL}_V$ is a representation, then for every $R \in \text{Alg}_k$, the group $G(R)$ acts on $V \otimes R$ on the left. The action extends to $\text{Sym}(V)$, and because Spm is contravariant, G acts on $\text{Spm} \text{Sym}(V)$ on the right. Equivalently, the comodule map on V extends to a comodule-algebra map

$$\text{Sym}(V) \longrightarrow \text{Sym}(V) \otimes \mathcal{O}(G).$$

Definition 2.19. A representation V of G is *simple* if $V \neq 0$ and V has no nontrivial subrepresentations.

Corollary 2.20. Every simple representation of a linear algebraic group is finite-dimensional.

Proof. A nonzero vector is contained in a finite-dimensional subrepresentation by the preceding local-finiteness proposition. Simplicity forces that subrepresentation to be all of V . $\square$

Let $X = G$, and let

$$m : G \times G \longrightarrow G$$

be right multiplication. The induced map

$$m^\# : \mathcal{O}(G) \longrightarrow \mathcal{O}(G) \otimes \mathcal{O}(G)$$

gives $\mathcal{O}(G)$ its regular $\mathcal{O}(G)$ -comodule structure.

Definition 2.21. The representation

$$r_G : G \longrightarrow \mathrm{GL}_{\mathcal{O}(G)}$$

associated with the comodule $(\mathcal{O}(G), m^\#)$ is called the *regular representation* of G .

Theorem 2.22. The regular representation of G has a faithful finite-dimensional subrepresentation. In particular, the regular representation is faithful.

Proof. Let $V \subseteq \mathcal{O}(G)$ be a finite-dimensional subcomodule. The subcomodule condition is expressed by the commutative diagram

$$\begin{array}{ccc} V & \hookrightarrow & \mathcal{O}(G) \\ \rho \downarrow & & \downarrow m^\# \\ V \otimes \mathcal{O}(G) & \hookrightarrow & \mathcal{O}(G) \otimes \mathcal{O}(G). \end{array}$$

Choose a basis $f_1, \dots, f_n$ of V . Write

$$m^\#(f_j) = \sum_{i=1}^n f_i \otimes b_{ij}, \quad b_{ij} \in \mathcal{O}(G).$$

Then the corresponding representation

$$G \longrightarrow \mathrm{GL}_V \simeq \mathrm{GL}_n$$

has matrix $(b_{ij}(g))$ at g , and hence its comorphism is

$$\mathcal{O}(\mathrm{GL}_n) = k[T_{ij}, \det(T)^{-1}] \longrightarrow \mathcal{O}(G), \quad T_{ij} \longmapsto b_{ij}.$$

This representation is faithful exactly when this map is surjective, that is, when the b_{ij} generate $\mathcal{O}(G)$ as a k -algebra.

Applying $e^\# \otimes \text{id}$ to $m^\#(f_j) = \sum_i f_i \otimes b_{ij}$ gives

$$f_j = \sum_{i=1}^n f_i(e) b_{ij}.$$

Therefore it is enough to choose V so that

- (1) V generates $\mathcal{O}(G)$ as a k -algebra;
- (2) V is an $\mathcal{O}(G)$ -subcomodule;
- (3) $\dim_k V < \infty$.

Because $\mathcal{O}(G)$ is a finitely generated k -algebra, choose finitely many algebra generators. By local finiteness, each generator is contained in a finite-dimensional subcomodule. The sum of those finitely many subcomodules satisfies all three conditions. Hence the resulting $G \rightarrow \text{GL}_V$ is faithful. $\square$

Corollary 2.23. If G is a linear algebraic group, then G is isomorphic to an algebraic subgroup of GL_n for some n .

Proof. Take the faithful finite-dimensional subrepresentation furnished by the theorem. A faithful finite-dimensional representation of a linear algebraic group is a closed immersion. $\square$

2.5 Constructing all finite-dimensional representations

Definition 2.24. For a k -vector space V , the comodule

$$V \otimes \mathcal{O}(G)$$

with coaction

$$\text{id}_V \otimes m^\# : V \otimes \mathcal{O}(G) \longrightarrow V \otimes \mathcal{O}(G) \otimes \mathcal{O}(G)$$

is called the *free comodule on V* .

Remark. If $\dim_k V = n$ and a basis of V is fixed, then

$$V \otimes \mathcal{O}(G) \simeq \mathcal{O}(G)^n,$$

with the componentwise free-comodule structure.

Proposition 2.25. Every finite-dimensional $\mathcal{O}(G)$ -comodule V is a subcomodule of

$$(\mathcal{O}(G)^n, (m^\#)^n)$$

for $n = \dim_k V$.

Proof. The coaction

$$\rho : V \longrightarrow V \otimes \mathcal{O}(G)$$

is injective, because

$$(\text{id}_V \otimes e^\#) \circ \rho = \text{id}_V .$$

Moreover, the diagram

$$\begin{array}{ccc} V & \xrightarrow{\rho} & V \otimes \mathcal{O}(G) \\ \rho \downarrow & & \downarrow \text{id}_V \otimes m^\# \\ V \otimes \mathcal{O}(G) & \xrightarrow{\rho \otimes \text{id}} & V \otimes \mathcal{O}(G) \otimes \mathcal{O}(G) \end{array}$$

commutes. Hence ρ is a comodule morphism from V into the free comodule $V \otimes \mathcal{O}(G)$. After choosing a basis of V ,

$$V \otimes \mathcal{O}(G) \simeq \mathcal{O}(G)^n .$$

□

Theorem 2.26. Let

$$r : G \longrightarrow \text{GL}_V$$

be a faithful finite-dimensional representation of G . Then every finite-dimensional representation W of G can be constructed from V by taking tensor products, finite direct sums, duals, and subquotients.

Proof. **Step 1: reduction to $W \subseteq \mathcal{O}(G)$.** By the preceding proposition, we may embed

$$W \hookrightarrow \mathcal{O}(G)^n$$

for some n . Let

$$W_i = \text{im} \left(W \hookrightarrow \mathcal{O}(G)^n \xrightarrow{\text{pr}_i} \mathcal{O}(G) \right) .$$

Then each W_i is a finite-dimensional subcomodule of $\mathcal{O}(G)$, and the map

$$W \longrightarrow \bigoplus_i W_i$$

is injective. Hence it is enough to treat finite-dimensional subcomodules $W \subseteq \mathcal{O}(G)$.

Step 2: reduction to $G = \mathrm{GL}_V$. Since r is faithful, its comorphism

$$r^\# : \mathcal{O}(\mathrm{GL}_V) \longrightarrow \mathcal{O}(G)$$

is surjective.

If

$$\widetilde{W} \subseteq \mathcal{O}(\mathrm{GL}_V)$$

is a finite-dimensional $\mathcal{O}(\mathrm{GL}_V)$ -subcomodule, then the Hopf-algebra identity

$$m_G^\# \circ r^\# = (r^\# \otimes r^\#) \circ m_{\mathrm{GL}_V}^\#$$

shows that

$$r^\#(\widetilde{W}) \subseteq \mathcal{O}(G)$$

is an $\mathcal{O}(G)$ -subcomodule. Equivalently, one has the commutative diagram

$$\begin{array}{ccccc} \widetilde{W} & \hookrightarrow & \mathcal{O}(\mathrm{GL}_V) & \xrightarrow{m_{\mathrm{GL}_V}^\#} & \mathcal{O}(\mathrm{GL}_V) \otimes \mathcal{O}(\mathrm{GL}_V) \\ \downarrow r^\#|_{\widetilde{W}} & & \downarrow r^\# & & \downarrow r^\# \otimes r^\# \\ r^\#(\widetilde{W}) & \hookrightarrow & \mathcal{O}(G) & \xrightarrow{m_G^\#} & \mathcal{O}(G) \otimes \mathcal{O}(G). \end{array}$$

Conversely, given a finite-dimensional subcomodule $W \subseteq \mathcal{O}(G)$, choose lifts in $\mathcal{O}(\mathrm{GL}_V)$ of a basis of W . By local finiteness of the regular representation of GL_V , those lifts are contained in a finite-dimensional subcomodule

$$\widetilde{W} \subseteq \mathcal{O}(\mathrm{GL}_V)$$

such that

$$r^\#(\widetilde{W}) \supseteq W.$$

Thus it is enough to treat $G = \mathrm{GL}_V$ case. ⁵

□

⁵2026.9.21 stopped at here.

A Homework

A.1 Week 1

Exercise A.1. Show that the two definitions (algebraic geometry definition and functor definition) of algebraic groups coincide.

Proof. In the class we proved that for any algebraic group G in the algebraic geometry sense, it induces

$$\tilde{G} : \text{Alg}_k \longrightarrow \text{Grp}, \quad R \longmapsto G(R),$$

and $\tilde{G}$ is an algebraic group in the functor sense.

It suffices to show that, for any functor $\tilde{G} : \text{Alg}_k \rightarrow \text{Grp}$ such that $\tilde{G}$, seen as $\text{Alg}_k \rightarrow \text{Set}$, is representable by a k -scheme G , then G is an algebraic group over k in the algebraic geometry sense.

Lemma A.2. [GW20, Corollary 4.7] Let X and Y be S -schemes. Then it is equivalent to give the following data.

1. A morphism of S -schemes from X to Y .
2. For all S -schemes T a map $f(T) : X_S(T) \rightarrow Y_S(T)$ of sets which is functorial in T , i.e., for all morphisms $u : T' \rightarrow T$ of S -schemes the following diagram is commutative:

$$\begin{array}{ccc} X_S(T) & \xrightarrow{f(T)} & Y_S(T) \\ X_S(u) \downarrow & & \downarrow Y_S(u) \\ X_S(T') & \xrightarrow{f(T')} & Y_S(T'). \end{array}$$

3. For all affine S -schemes $T = \text{Spec } B$ a map $f(T) : X_S(T) \rightarrow Y_S(T)$ of sets which is functorial in B .

- **Construction of $G \times G \xrightarrow{m} G$.** By Lemma A.2, it is equivalent to give, for all $R \in \text{Alg}_k$, a map

$$(G \times G)(R) \xrightarrow{m(R)} G(R)$$

and, for all k -algebra homomorphisms $R \rightarrow T$, to have the following diagram commute:

$$\begin{array}{ccc} (G \times G)(R) & \xrightarrow{m(R)} & G(R) \\ \downarrow & & \downarrow \\ (G \times G)(T) & \xrightarrow{m(T)} & G(T). \end{array}$$

Because it is well-known that $(G \times G)(R) = G(R) \times G(R)$ and $G(R) \cong \tilde{G}(R)$ is a group, $m(R)$ can just be the multiplication of $\tilde{G}(R)$. Because the vertical maps are group homomorphisms, hence it commutes.

- **Construction of $e : \operatorname{Spec} k \rightarrow G$.** It just corresponds to $G(k) \cong \tilde{G}(k)$, namely the element $1_{\tilde{G}(k)}$.
- **Construction of $i : G \rightarrow G$.** By Lemma A.2, it is equivalent to give, for all $R \in \operatorname{Alg}_k$, a map

$$G(R) \xrightarrow{i(R)} G(R)$$

and, for all k -algebra homomorphisms $R \rightarrow T$, to have the following diagram commute:

$$\begin{array}{ccc} G(R) & \xrightarrow{i(R)} & G(R) \\ \downarrow & & \downarrow \\ G(T) & \xrightarrow{i(T)} & G(T). \end{array}$$

Since $G(R) \cong \tilde{G}(R)$ is a group, $i(R)$ can just be the inverse map of $\tilde{G}(R)$. Because the vertical maps are group homomorphisms, hence it commutes.

- **Associativity.** To prove that

$$\begin{array}{ccc} G \times G \times G & \xrightarrow{(m, \operatorname{id}_G)} & G \times G \\ (\operatorname{id}_G, m) \downarrow & & \downarrow m \\ G \times G & \xrightarrow{m} & G \end{array}$$

commutes, i.e.,

$$G \times G \times G \xrightarrow{(m, \operatorname{id}_G)} G \times G \xrightarrow{m} G$$

and

$$G \times G \times G \xrightarrow{(\operatorname{id}_G, m)} G \times G \xrightarrow{m} G$$

give the same morphism, by Lemma A.2 it suffices to show that for every $R \in \text{Alg}_k$,

$$(G \times G \times G)(R) \xrightarrow{(m(R), \text{id}_{G(R)})} (G \times G)(R) \xrightarrow{m(R)} G(R)$$

and

$$(G \times G \times G)(R) \xrightarrow{(\text{id}_{G(R)}, m(R))} (G \times G)(R) \xrightarrow{m(R)} G(R)$$

give the same map. It is obvious by $G(R) \cong \tilde{G}(R)$ being a group and the construction of $m(R)$, i.e., it is just the associativity on $\tilde{G}(R)$.

- **Existence of unit.** To show

$$\begin{array}{ccc} G \cong G \times \text{Spec } k & \xrightarrow{(\text{id}_G, e)} & G \times G \\ & \searrow \text{id}_G & \downarrow m \\ & & G \end{array}$$

commutes, it suffices to show for every $R \in \text{Alg}_k$ that

$$\begin{array}{ccc} G(R) & \xrightarrow{g \mapsto (g, 1)} & G(R) \times G(R) \\ & \searrow \text{id}_{G(R)} & \downarrow m(R) \\ & & G(R) \end{array}$$

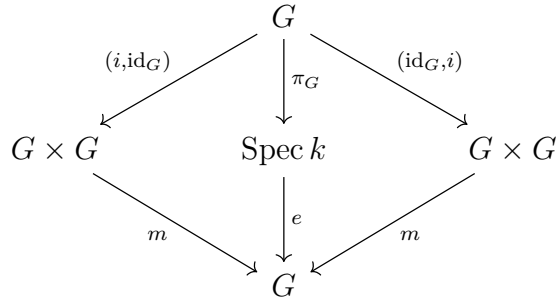
commutes, obviously.

Similarly,

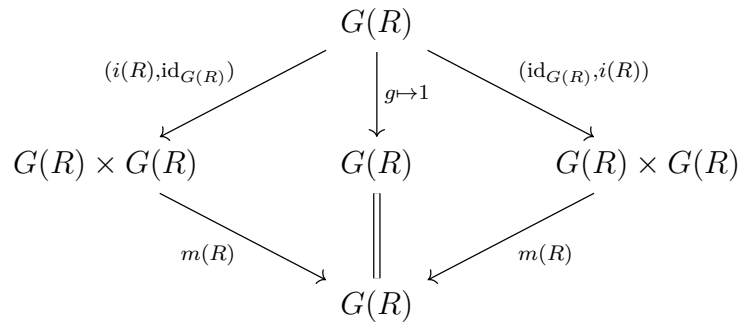
$$\begin{array}{ccc} G & \xrightarrow{(e, \text{id}_G)} & G \times G \\ & \searrow \text{id}_G & \downarrow m \\ & & G \end{array}$$

commutes can be proved similarly.

- **Inverse property.** To show the following diagram commutes,



it suffices to show, for every $R \in \text{Alg}_k$, the following diagram of maps commutes:



namely $g \mapsto (g^{-1}, g) \mapsto 1$ and $g \mapsto (g, g^{-1}) \mapsto 1$, obviously.

□

Exercise A.3. (a) Show that GL_n , B_n and U_n are linear algebraic groups.

(b) Show that U_n and B_n are both algebraic subgroups of GL_n .

(c) write down $m^\#$, $i^\#$ and $e^\#$ for GL_2 .

(d) Define the functor $\text{SL}_n : \text{Alg}_k \rightarrow (\text{gps})$ such that

$$\text{SL}_n(R) = \{A \in \text{GL}_n(R) \mid \det(A) = 1\}.$$

Show that it's a linear algebraic group, called special linear group.

(e) Compute the dimension of all these algebraic groups GL_n , SL_n , B_n and U_n . Here dimension of an algebraic group is defined to be the dimension of the underlying scheme over k .

Proof of (a). **Claim.** See GL_n, B_n, U_n as functors from Alg_k to Grp as

$$\begin{aligned}\mathrm{GL}_n(R) &= \{(x_{ij}) \in M_n(R) \mid \det(x_{ij}) \in R^\times\}, \\ B_n(R) &= \{(x_{ij}) \in \mathrm{GL}_n(R) \mid x_{ij} = 0 \text{ for } i > j\}, \\ U_n(R) &= \{(x_{ij}) \in B_n(R) \mid x_{ii} = 1 \text{ for all } i\}.\end{aligned}$$

$\mathrm{GL}_n(B_n, U_n)$ is represented by affine $\mathrm{GL}_{n,k}(B_{n,k}, U_{n,k})$. In fact,

$$\mathrm{GL}_{n,k} = \mathrm{Spm} k[X_{ij}]_{\det(X_{ij})},$$

where $\det(X_{ij})$ is a polynomial in $k[X_{ij}]$ and the subscript denotes localization,

$$B_{n,k} = \mathrm{Spm} (k[X_{ij}]_{\det(X_{ij})} / (X_{ij})_{i>j}) = \mathrm{Spm} ((k[X_{ij}]_{i \leq j})_{X_{11}X_{22}\cdots X_{nn}}),$$

and

$$U_{n,k} = \mathrm{Spm} \left(\frac{(k[X_{ij}]_{i \leq j})_{X_{11}X_{22}\cdots X_{nn}}}{(X_{11} - 1, X_{22} - 1, \dots, X_{nn} - 1)} \right) \cong \mathrm{Spm}(k[X_{ij}]_{i < j}).$$

Reason: it suffices to show that

$$\mathrm{GL}_n(R) \cong \mathrm{Hom}_{\mathrm{Sch}}(\mathrm{Spm} R, \mathrm{GL}_{n,k}) \cong \mathrm{Hom}_{\mathrm{Alg}_k}(k[X_{ij}]_{\det(X_{ij})}, R), \quad (1)$$

$$B_n(R) \cong \mathrm{Hom}_{\mathrm{Alg}_k}((k[X_{ij}]_{i \leq j})_{X_{11}\cdots X_{nn}}, R), \quad (2)$$

$$U_n(R) \cong \mathrm{Hom}_{\mathrm{Alg}_k}(k[X_{ij}]_{i < j}, R). \quad (3)$$

Notice that

$$M_n(R) \cong \mathrm{Hom}_{\mathrm{Alg}_k}(k[X_{ij}], R), \quad (r_{ij}) \longleftrightarrow (X_{ij} \mapsto r_{ij}).$$

Thus (1) and (2) follow from the universal property of localization:

$$\begin{array}{ccc} k[X_{ij}] & \xrightarrow{X_{ij} \mapsto r_{ij}} & R \\ \downarrow & \nearrow \text{dashed} & \\ k[X_{ij}]_{\det(X_{ij})} & & \end{array} \quad \text{with } \det(r_{ij}) \in R^\times,$$

and (3) is direct. □

Proof of (b). By our construction in (a),

$$U_n \hookrightarrow B_n \hookrightarrow \mathrm{GL}_n.$$

Show $U_n \hookrightarrow B_n \hookrightarrow \mathrm{GL}_n$ are homomorphisms of algebraic groups in the functorial sense; i.e.,

for every $R \in \text{Alg}_k$,

$$U_n(R) \hookrightarrow B_n(R) \hookrightarrow \text{GL}_n(R)$$

are homomorphisms of groups (inclusion). It is natural.

But there is a detail: the inclusion $B_n(R) \rightarrow \text{GL}_n(R)$ for any $R \in \text{Alg}_k$, which is obviously functorial on Alg_k , just corresponds to the natural closed immersion $B_n \hookrightarrow \text{GL}_n$ in Lemma A.2.

Reason: The natural closed immersion $B_n \hookrightarrow \text{GL}_n$ corresponds to the natural quotient map

$$S \longrightarrow S/I, \quad S = k[X_{ij}]_{\det(X_{ij})}, \quad I \text{ generated by } (X_{ij})_{i>j}.$$

For every $R \in \text{Alg}_k$, the induced

$$\text{Mor}(\text{Spm } R, B_n) \longrightarrow \text{Mor}(\text{Spm } R, \text{GL}_n)$$

corresponds to

$$\begin{array}{ccc} S & \xrightarrow{\text{natural}} & S/I \\ & \searrow & \downarrow \\ & & R, \end{array}$$

and corresponds to $B_n(R) \rightarrow \text{GL}_n(R)$ by

$$(r_{ij}) \longmapsto (r_{ij}),$$

that is,

$$\left(\begin{array}{c} X_{ij} \mapsto r_{ij} \\ S/I \rightarrow R \end{array} \right) \longmapsto \left(\begin{array}{c} X_{ij} \mapsto r_{ij} \\ S \rightarrow R \end{array} \right),$$

which is just the natural inclusion. □

Solution of (c).

$$m^\# : k[x, y, z, w]_{(xw-yz)} \longrightarrow k[x, y, z, w]_{(xw-yz)} \otimes k[x, y, z, w]_{(xw-yz)}.$$

See an element in $k[x, y, z, w]_{(xw-yz)}$ as a function on GL_2 . For example, if $m^\#(x) = \sum_i f_i \otimes g_i$, then for

$$\begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix}, \quad \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} \in \text{GL}_2,$$

we have

$$a_1a_2 + b_1c_2 = x \left(\begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} \right) = \sum_i f_i \left(\begin{pmatrix} a_1 & b_1 \\ c_1 & d_1 \end{pmatrix} \right) g_i \left(\begin{pmatrix} a_2 & b_2 \\ c_2 & d_2 \end{pmatrix} \right).$$

Thus

$$m^\#(x) = x \otimes x + y \otimes z.$$

Similarly,

$$m^\#(y) = x \otimes y + y \otimes w,$$

$$m^\#(z) = z \otimes x + w \otimes z,$$

$$m^\#(w) = z \otimes y + w \otimes w.$$

Also

$$e^\# : k[x, y, z, w]_{(xw-yz)} \longrightarrow k, \quad x, w \longmapsto 1, \quad y, z \longmapsto 0,$$

corresponding to $\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$, and

$$i^\# : k[x, y, z, w]_{(xw-yz)} \longrightarrow k[x, y, z, w]_{(xw-yz)},$$

with

$$\begin{aligned} x &\longmapsto (xw - yz)^{-1}w, \\ y &\longmapsto -(xw - yz)^{-1}y, \\ z &\longmapsto -(xw - yz)^{-1}z, \\ w &\longmapsto (xw - yz)^{-1}x, \end{aligned} \quad \text{corresponding to } A \longmapsto A^{-1}.$$

□

Proof of (d). $\mathrm{SL}_n(R)$ are groups for any $R \in \mathrm{Alg}_k$. It suffices to show SL_n , as a functor from Alg_k to Set , is represented by

$$\mathrm{SL}_{n,k} := \mathrm{Spm} \left(\frac{k[X_{ij}]_{\det(X_{ij})}}{(\det(X_{ij}) - 1)} \right) = \mathrm{Spm} \left(\frac{k[X_{ij}]}{(\det(X_{ij}) - 1)} \right).$$

It suffices to show that, for any $R \in \mathrm{Alg}_k$,

$$\mathrm{SL}_n(R) \cong \mathrm{Hom}_{\mathrm{Sch}}(\mathrm{Spm} R, \mathrm{SL}_n) \cong \mathrm{Hom}_{\mathrm{Alg}_k} \left(\frac{k[X_{ij}]}{(\det(X_{ij}) - 1)}, R \right),$$

and this is satisfied by

$$(r_{ij}) \longleftrightarrow (X_{ij} \mapsto r_{ij}).$$

□

Solution of (e). GL_n is an open subscheme of $\mathrm{Spm} k[X_{ij}]$, hence

$$\dim \mathrm{GL}_n = n^2.$$

SL_n is a hypersurface of $\mathrm{Spm} k[X_{ij}]$, hence

$$\dim \mathrm{SL}_n = n^2 - 1.$$

B_n is cut out of GL_n by $\frac{n(n-1)}{2}$ functions X_{ij} , $i > j$. Hence

$$\dim B_n = n^2 - \frac{n(n-1)}{2} = \frac{n(n+1)}{2}.$$

Finally,

$$U_n \cong \mathrm{Spm} k[X_{ij}]_{i < j}, \quad \dim U_n = \frac{n(n-1)}{2}.$$

□

Exercise A.4. For $C \in \mathrm{GL}_n(k)$, consider the functor $G_C : \mathrm{Alg}_k \rightarrow (\mathrm{gps})$ with

$$G_C(R) = \{A \in \mathrm{GL}_n(R) \mid A^t C A = C\}.$$

(a). Show that G_C is a linear algebraic group.

(b). When $n = 2m$ is even and $C = \begin{pmatrix} 0 & I_m \\ -I_m & 0 \end{pmatrix}$, then G_C is called the symplectic group denoted by Sp_n . Compute its dimension.

(c). For $\mathrm{char}(k) \neq 2$, when $C = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & I_m \\ 0 & I_m & 0 \end{pmatrix}$ (resp. $C = \begin{pmatrix} 0 & I_m \\ I_m & 0 \end{pmatrix}$) if $n = 2m + 1$ odd (resp. $n = 2m$ even), then G_C is called the orthogonal group denoted by O_n . Compute its dimension.

Proof of (a). $G_C(R)$ is a group for all $R \in \mathrm{Alg}_k$. Write

$$C = (c_{ij}), \quad c_{ij} \in k.$$

Then

$$(X_{ij})^t(c_{ij})(X_{ij}) - (c_{ij}) =: (f_{ij}(X_{11}, \dots, X_{nn})), \quad f_{ij} \in k[X_{ij}].$$

Claim. $G_C(-)$ is represented by

$$\mathrm{Spm} \frac{k[X_{ij}]}{(f_{ij})},$$

which is affine.

According to the definition, at first it should be

$$\mathrm{Spm} \left(k[X_{ij}]_{\det(X_{ij})} / (f_{ij}) \right),$$

but $\det(X_{ij})$ can be removed because $A^t C A = C$ and $C \in \mathrm{GL}_n$ imply $A \in \mathrm{GL}_n$.

Reason: It suffices to show, for any $R \in \mathrm{Alg}_k$,

$$G_C(R) \cong \mathrm{Hom}_{\mathrm{Sch}} \left(\mathrm{Spm} R, \mathrm{Spm} \frac{k[X_{ij}]}{(f_{ij})} \right) \cong \mathrm{Hom}_{\mathrm{Alg}_k} \left(\frac{k[X_{ij}]}{(f_{ij})}, R \right),$$

and it is satisfied by

$$(r_{ij}) \longleftrightarrow (X_{ij} \mapsto r_{ij}).$$

□

Solution of (b). Notice that, because C is anti-symmetric,

$$f_{ij} = -f_{ji}.$$

Therefore $(f_{ij}) \subseteq k[X_{ij}]$ can be generated by at most $\frac{n(n-1)}{2}$ polynomials. Hence

$$\dim \mathrm{Sp}_n \geq n^2 - \frac{n(n-1)}{2} = \frac{n(n+1)}{2}.$$

We want to compute the corank of the Jacobian to get the upper bound of $\dim \mathrm{Sp}_n$. By [Vak25, 13.1.M, 13.1.N], for every $p \in \mathrm{Sp}_n$,

$$\begin{aligned} \dim_{\mathrm{Sp}_n} p &\leq \mathrm{corank} \left(\left(\frac{\partial f_{ij}}{\partial X_{kl}}(p) \right)_{i,j,k,l \leq n} \right) \\ &= \dim_{\kappa(p)} \left\{ B = (b_{kl}) \in M_{n \times n}(\kappa(p)) \mid \forall i, j, \sum_{k,l} \frac{\partial f_{ij}}{\partial X_{kl}}(p) \cdot b_{kl} = 0 \right\}. \end{aligned}$$

By

$$f_{ij} = \sum_{r,s} X_{ri} C_{rs} X_{sj} - c_{ij},$$

we have, for every k, l ,

$$\frac{\partial f_{ij}}{\partial X_{kl}} = \delta_{il} \sum_s c_{ks} X_{sj} + \delta_{jl} \sum_r X_{ri} c_{rk}.$$

Thus, for every i, j ,

$$\begin{aligned} \sum_{k,l} \frac{\partial f_{ij}}{\partial X_{kl}}(p) \cdot b_{kl} &= \sum_{k,s} b_{ki} c_{ks} X_{sj}(p) + \sum_{r,k} X_{ri}(p) c_{rk} b_{kj} \\ &= (B^t C A(p) + A(p)^t C B)_{ij}. \end{aligned}$$

Therefore

$$\begin{aligned} \dim_{\text{Sp}_n} p &\leq \dim_{\kappa(p)} \{B \in M_{n \times n}(\kappa(p)) \mid B^t C A(p) + A(p)^t C B = 0\} \\ &= \dim_{\kappa(p)} \{B \in M_{n \times n}(\kappa(p)) \mid B^t C A(p) = (B^t C A(p))^t\} \quad \text{by } C = \begin{pmatrix} 0 & I_m \\ -I_m & 0 \end{pmatrix} \\ &= \dim_{\kappa(p)} \{Y \in M_{n \times n}(\kappa(p)) \mid Y = Y^t\} \quad \text{by } C, A(p) \in \text{GL}_n(\kappa(p)) \\ &= \frac{n(n+1)}{2}. \end{aligned}$$

Hence

$$\dim \text{Sp}_n = \frac{n(n+1)}{2}.$$

□

Solution of (c). Because C is symmetric,

$$f_{ij} = f_{ji}.$$

Therefore $(f_{ij}) \subseteq k[X_{ij}]$ can be generated by at most $\frac{n(n+1)}{2}$ polynomials. Hence

$$\dim \text{O}_n \geq n^2 - \frac{n(n+1)}{2} = \frac{n(n-1)}{2}.$$

By a similar discussion as in (b), for every $p \in \text{O}_n$,

$$\begin{aligned} \dim_{\text{O}_n} p &\leq \dim_{\kappa(p)} \{B \in M_{n \times n}(\kappa(p)) \mid B^t C A(p) + A(p)^t C B = 0\} \\ &= \dim_{\kappa(p)} \{B \in M_{n \times n}(\kappa(p)) \mid B^t C A(p) = -(B^t C A(p))^t\} \quad \text{by } C = C^t \text{ and } \text{char } k \neq 2 \\ &= \dim_{\kappa(p)} \{Y \in M_{n \times n}(\kappa(p)) \mid Y = -Y^t\} \quad \text{by } C, A(p) \in \text{GL}_n(\kappa(p)) \\ &= \frac{n(n-1)}{2}. \end{aligned}$$

Hence

$$\dim \mathcal{O}_n = \frac{n(n-1)}{2}.$$

□

Exercise A.5. Show that the sequence of homomorphism of algebraic groups

$$e \rightarrow \mathrm{SL}_n \rightarrow \mathrm{GL}_n \xrightarrow{\det} \mathbb{G}_m \rightarrow e$$

is exact, here the first map is the natural inclusion and the second one is the determinant map.

Proof. • $\mathrm{SL}_n \cong \mathrm{Ker}(\det)$. For every $R \in \mathrm{Alg}_k$,

$$K(R) := \ker(\mathrm{GL}_n(R) \xrightarrow{\det} \mathbb{G}_m(R) = R^\times)$$

is a group. By definition, $K(R)$ is represented by $\mathrm{Ker}(\mathrm{GL}_n \xrightarrow{\det} \mathbb{G}_m)$. Also, $K(R) = \mathrm{SL}_n(R)$ is represented by SL_n .

- $\mathrm{GL}_n \xrightarrow{\det} \mathbb{G}_m$ is faithfully flat, i.e. flat and surjective by [Vak25, The paragraph after 24.5.B].

Surjectivity. It suffices to find a section $s : \mathbb{G}_m \rightarrow \mathrm{GL}_n$. It suffices to find a k -algebra homomorphism

$$s^\# : k[X_{ij}]_{\det(X_{ij})} \longrightarrow k[t, t^{-1}]$$

such that, with

$$\det^\# : k[t, t^{-1}] \longrightarrow k[X_{ij}]_{\det(X_{ij})}, \quad t \longmapsto \det(X_{ij}),$$

we have

$$s^\# \circ \det^\# = \mathrm{id}_{k[t, t^{-1}]}.$$

Consider $s^\#$ given by

$$X_{11} \longmapsto t, \quad X_{ii} \longmapsto 1 \text{ for } i > 1, \quad X_{ij} \longmapsto 0 \text{ for } i \neq j.$$

So

$$\begin{array}{ccccc} \mathbb{G}_m & \xrightarrow{s} & \mathrm{GL}_n & \xrightarrow{\det} & \mathbb{G}_m, \\ & \searrow & & \nearrow & \\ & & \mathrm{id} & & \end{array}$$

and $\det$ is surjective.

Flatness. With

$$k[t, t^{-1}] \xrightarrow{\det^\#} k[X_{ij}]_{\det(X_{ij})}$$

injective by the above choice of the section and the fact that $k[X_{ij}]_{\det(X_{ij})}$ is integral, $k[X_{ij}]_{\det(X_{ij})}$ is torsion-free as a $k[t, t^{-1}]$ -module. The ring $k[t, t^{-1}]$ is a localization of a PID, hence is a PID. A torsion-free algebra over a PID is flat over it. □

A.2 Week 2

Exercise A.6. The natural isomorphism of set valued functors

$$\mathrm{SL}_n(R) \times \mathbb{G}_m(R) \longrightarrow \mathrm{GL}_n(R), \quad (A, r) \longmapsto A \cdot \mathrm{diag}(r, 1, \dots, 1)$$

defines an isomorphism of k -algebras

$$\mathcal{O}(\mathrm{GL}_n) \simeq \mathcal{O}(\mathrm{SL}_n) \otimes \mathcal{O}(\mathbb{G}_m).$$

(a) Show that SL_n is connected.

(b) Please describe the map

$$\mathcal{O}(\mathrm{SL}_n) \longrightarrow \mathcal{O}(\mathrm{GL}_n)$$

from the previous isomorphism.

Proof. (a) By the given isomorphism and Yoneda,

$$\mathrm{SL}_n \times \mathbb{G}_m \simeq \mathrm{GL}_n.$$

Also

$$\mathrm{GL}_n \simeq \mathrm{Spm} k[X_{ij}]_\Delta, \quad \Delta = \det(X_{ij}).$$

Since $k[X_{ij}]_\Delta$ is an integral domain, GL_n is irreducible, hence connected. Consider

$$\mathrm{GL}_n \xrightarrow{\sim} \mathrm{SL}_n \times \mathbb{G}_m \xrightarrow{\mathrm{pr}_1} \mathrm{SL}_n.$$

It is surjective. The image of a connected space is connected, so SL_n is connected.

(b) The map in question is induced by

$$\mathrm{GL}_n \xrightarrow{\sim} \mathrm{SL}_n \times \mathbb{G}_m \xrightarrow{\mathrm{pr}_1} \mathrm{SL}_n.$$

On coordinate rings it is

$$\mathcal{O}(\mathrm{SL}_n) \longrightarrow \mathcal{O}(\mathrm{SL}_n) \otimes \mathcal{O}(\mathbb{G}_m) \xrightarrow{\sim} \mathcal{O}(\mathrm{GL}_n), \quad f \longmapsto f \otimes 1.$$

Write

$$\mathcal{O}(\mathrm{SL}_n) = \frac{k[X_{ij}]}{(\Delta - 1)}, \quad \mathcal{O}(\mathbb{G}_m) = k[T, T^{-1}], \quad \mathcal{O}(\mathrm{GL}_n) = k[X_{ij}]_{\Delta},$$

and let $\overline{X}_{ij}$ denote the image of X_{ij} in $\mathcal{O}(\mathrm{SL}_n)$. Under the given functorial isomorphism,

$$X_{i1} \longmapsto \overline{X}_{i1} \otimes T, \quad X_{ij} \longmapsto \overline{X}_{ij} \otimes 1 \quad (j \neq 1), \quad \Delta \longmapsto 1 \otimes T.$$

Therefore

$$\boxed{\overline{X}_{i1} \longmapsto \frac{X_{i1}}{\Delta}, \quad \overline{X}_{ij} \longmapsto X_{ij} \quad (j \neq 1).}$$

□

Exercise A.7. Let $\mathrm{char}(k) \neq 2$. Let $O' = G_C$ with $C = I_n$ (notation from Week 1 Ex. 3).

(a) Show that

$$O_{n, \overline{k}} \simeq O'_{n, \overline{k}}$$

as algebraic groups over $\overline{k}$.

(b) Show that the determinant map

$$\det : O_n \longrightarrow \mu_2$$

is a homomorphism of algebraic groups, where

$$\mu_2 = \mathrm{Ker}(\mathbb{G}_m \longrightarrow \mathbb{G}_m, x \longmapsto x^2).$$

Define

$$\mathrm{SO}_n := \mathrm{Ker}(\det : O_n \rightarrow \mu_2).$$

And similarly define

$$\mathrm{SO}'_n := \mathrm{Ker}(\det : O'_n \rightarrow \mu_2).$$

(c) For each k -algebra R , let $V(R)$ denote the set of skew-symmetric matrices, i.e.

the matrices $A \in M_n(R)$ such that $A^t = -A$. Show that the functor

$$R \longmapsto V(R)$$

is represented by a finitely generated k -algebra C , and that C is an integral domain.

(d) Show that

$$A \longmapsto (I_n - A)(I_n + A)^{-1}$$

defines a bijection from a nonempty open subset of $\mathrm{SO}'_n(\bar{k})$ onto an open subset of $V(\bar{k})$, with partial inverse

$$B \longmapsto (I_n - B)(I_n + B)^{-1}.$$

(e) Deduce that $\mathrm{SO}'_{n,\bar{k}}$ is connected.

(f) Deduce that SO_n is connected.

Proof. Let J be the symmetric matrix defining O_n in Week 1, namely

$$J = \begin{cases} \begin{pmatrix} 0 & I_m \\ I_m & 0 \end{pmatrix}, & n = 2m, \\ \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & I_m \\ 0 & I_m & 0 \end{pmatrix}, & n = 2m + 1. \end{cases}$$

(a) Choose $\alpha, \beta \in \bar{k}$ such that

$$\alpha^2 = 2, \quad \beta^2 = -1.$$

For the 2×2 hyperbolic block we have

$$\begin{pmatrix} \frac{1}{\alpha} & \frac{1}{\alpha} \\ \frac{\beta}{\alpha} & -\frac{\beta}{\alpha} \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \frac{1}{\alpha} & \frac{\beta}{\alpha} \\ \frac{1}{\alpha} & -\frac{\beta}{\alpha} \end{pmatrix} = I_2.$$

Applying this to the hyperbolic blocks of J gives $Q \in \mathrm{GL}_n(\bar{k})$ such that

$$Q^t J Q = I_n.$$

Hence, for every $R \in \text{Alg}_{\bar{k}}$,

$$O_{n,\bar{k}}(R) = \{A \in \text{GL}_n(R) \mid A^t J A = J\}$$

is naturally identified with

$$\{B \in \text{GL}_n(R) \mid B^t I_n B = I_n\} = O'_{n,\bar{k}}(R)$$

by

$$A \longmapsto Q^{-1} A Q.$$

Indeed, $Q^t J Q = I_n$ implies $J = Q^{-t} Q^{-1}$, and therefore

$$(Q^{-1} A Q)^t (Q^{-1} A Q) = Q^t A^t J A Q = Q^t J Q = I_n.$$

The inverse is $B \mapsto Q B Q^{-1}$. This is functorial in R , so by Yoneda

$$O_{n,\bar{k}} \simeq O'_{n,\bar{k}}.$$

(b) It is enough to show that for every $R \in \text{Alg}_k$,

$$\det(R) : O_n(R) \longrightarrow \mu_2(R)$$

is a group homomorphism. We have

$$\mu_2(R) = \text{Ker}\left(\mathbb{G}_m(R) \xrightarrow{r \mapsto r^2} \mathbb{G}_m(R)\right) = \{r \in R^\times \mid r^2 = 1\}.$$

For $A \in O_n(R)$, from

$$A^t J A = J$$

we get

$$\det(A)^2 \det(J) = \det(J).$$

Since $\det(J) \in k^\times$, it follows that

$$\det(A)^2 = 1.$$

Thus the image of $O_n(R)$ under determinant is contained in $\mu_2(R)$. Moreover,

$$O_n(R) \xrightarrow{\det} \mathbb{G}_m(R)$$

is a group homomorphism. Hence

$$O_n(R) \longrightarrow \mu_2(R)$$

is a group homomorphism for every R , and therefore

$$\det : O_n \longrightarrow \mu_2$$

is a homomorphism of algebraic groups.

(c) We claim that V is represented by

$$\operatorname{Spm} \frac{k[X_{ij} \mid 1 \leq i, j \leq n]}{(X_{ij} + X_{ji} \mid 1 \leq i, j \leq n)}.$$

Indeed, for every $R \in \operatorname{Alg}_k$,

$$\begin{aligned} \operatorname{Hom}_{k\text{-alg}} \left(\frac{k[X_{ij}]}{(X_{ij} + X_{ji})}, R \right) &\simeq \{f \in \operatorname{Hom}_{k\text{-alg}}(k[X_{ij}], R) \mid f(X_{ij}) + f(X_{ji}) = 0 \ \forall i, j\} \\ &\simeq \{(r_{ij}) \in M_n(R) \mid r_{ij} + r_{ji} = 0 \ \forall i, j\} \\ &= V(R). \end{aligned}$$

Thus the claim follows by Yoneda. Since $\operatorname{char}(k) \neq 2$, the relations imply

$$X_{ii} = 0, \quad X_{ji} = -X_{ij},$$

and hence

$$C := \frac{k[X_{ij}]}{(X_{ij} + X_{ji})} \simeq k[X_{ij} \mid i < j].$$

Therefore C is finitely generated and is an integral domain.

(d) Consider, as classical varieties,

$$U = \{A \in \operatorname{SO}'_n(\bar{k}) \mid \det(I_n + A) \neq 0\}$$

and

$$W = \{B \in V(\bar{k}) \mid \det(I_n + B) \neq 0\}.$$

These are open, and they are nonempty since $I_n \in U$ and $0 \in W$. We claim that

$$U \xrightarrow{\sim} W, \quad A \longmapsto (I_n - A)(I_n + A)^{-1},$$

with inverse

$$B \longmapsto (I_n - B)(I_n + B)^{-1}.$$

First, it is enough to check the bijection set-theoretically. Indeed, the entries of $(I_n - Y)(I_n + Y)^{-1}$ belong to the localization in which $\det(I_n + Y)$ is inverted, so they define the corresponding homomorphism of localized $\bar{k}$ -algebras. The other direction is defined similarly. Thus, once the two formulas are checked to be inverse on points, they give inverse morphisms of the two open subschemes.

Let $A \in U$ and put

$$B = (I_n - A)(I_n + A)^{-1}.$$

Since $A^t A = I_n$, we have $A^t = A^{-1}$, and

$$\begin{aligned} B^t &= (I_n + A^t)^{-1}(I_n - A^t) \\ &= (I_n + A^{-1})^{-1}(I_n - A^{-1}) \\ &= (A + I_n)^{-1}(A - I_n) \\ &= (A - I_n)(A + I_n)^{-1} \\ &= -B. \end{aligned}$$

Hence $B \in V(\bar{k})$. Moreover,

$$I_n + B = 2(I_n + A)^{-1},$$

so $\det(I_n + B) \neq 0$, and hence $B \in W$.

Conversely, let $B \in W$ and put

$$A = (I_n - B)(I_n + B)^{-1}.$$

Since $B^t = -B$,

$$\begin{aligned} A^t &= (I_n + B^t)^{-1}(I_n - B^t) \\ &= (I_n - B)^{-1}(I_n + B) \\ &= A^{-1}. \end{aligned}$$

Also

$$\det(A) = \frac{\det(I_n - B)}{\det(I_n + B)} = \frac{\det((I_n - B)^t)}{\det(I_n + B)} = \frac{\det(I_n + B)}{\det(I_n + B)} = 1.$$

Thus $A \in \mathrm{SO}'_n(\bar{k})$. Moreover,

$$I_n + A = 2(I_n + B)^{-1},$$

so $A \in U$.

Finally, if $B = (I_n - A)(I_n + A)^{-1}$, then

$$I_n - B = 2A(I_n + A)^{-1}, \quad I_n + B = 2(I_n + A)^{-1},$$

and hence

$$(I_n - B)(I_n + B)^{-1} = A.$$

The other composition is the same. Therefore $U \simeq W$.

(e) We use the Cartan–Dieudonné theorem: for an n -dimensional nondegenerate symmetric bilinear space over a field of characteristic not 2, every element of the orthogonal group is a composition of at most n reflections.

For a nonisotropic vector v , write

$$S_v(x) = x - \frac{2\langle x, v \rangle}{\langle v, v \rangle} v.$$

Its eigenvalues are $-1, 1, \dots, 1$, so $\det(S_v) = -1$.

Let U be the open subset from part (d). We claim that for any nonisotropic vectors u, v ,

$$S_u S_v \in U(\bar{k})^2.$$

Because $\bar{k}$ is infinite, we can choose a nonisotropic vector w such that

$$\langle u, w \rangle \neq 0, \quad \langle v, w \rangle \neq 0.$$

Indeed, we choose w outside the union of the three proper closed subsets cut out by

$$\langle w, w \rangle, \quad \langle u, w \rangle, \quad \langle v, w \rangle.$$

We show that $S_u S_w \in U(\bar{k})$. Suppose, to the contrary, that -1 is an eigenvalue of $S_u S_w$. Choose $x \neq 0$ with

$$S_u S_w x = -x.$$

Since $S_u^2 = 1$, this gives

$$S_w x = -S_u x.$$

Using the definitions of the two reflections,

$$x - \frac{2\langle x, w \rangle}{\langle w, w \rangle} w = -x + \frac{2\langle x, u \rangle}{\langle u, u \rangle} u,$$

so

$$x = \frac{\langle x, u \rangle}{\langle u, u \rangle} u + \frac{\langle x, w \rangle}{\langle w, w \rangle} w.$$

Pairing this equality first with u and then with w gives

$$\frac{\langle x, w \rangle}{\langle w, w \rangle} \langle u, w \rangle = 0, \quad \frac{\langle x, u \rangle}{\langle u, u \rangle} \langle u, w \rangle = 0.$$

Since $x \neq 0$, the two coefficients on the left cannot both be zero. Hence $\langle u, w \rangle = 0$, a contradiction. Therefore -1 is not an eigenvalue of $S_u S_w$, so

$$\det(I_n + S_u S_w) \neq 0.$$

Since $\det(S_u S_w) = 1$, we have $S_u S_w \in U(\bar{k})$. Similarly, $S_w S_v \in U(\bar{k})$. Hence

$$S_u S_v = (S_u S_w)(S_w S_v) \in U(\bar{k})^2.$$

Now let $A \in \mathrm{SO}'_n(\bar{k})$. By Cartan–Dieudonné,

$$A = S_{v_1} \cdots S_{v_\ell}.$$

Since

$$1 = \det(A) = (-1)^\ell,$$

the integer ℓ is even. Pairing the reflections and applying the claim shows that

$$A \in \langle U(\bar{k}) \rangle.$$

Therefore

$$\mathrm{SO}'_n(\bar{k}) = \langle U(\bar{k}) \rangle.$$

By part **(d)**, U is isomorphic to the nonempty open subset W of V . By part **(c)**, V is irreducible. Hence U is irreducible, and therefore connected. Since $I_n \in U$, we have

$$U \subseteq (\mathrm{SO}'_{n,\bar{k}})^\circ.$$

The identity component is a subgroup, so it contains the subgroup generated by $U(\bar{k})$. As this subgroup is all of $\mathrm{SO}'_n(\bar{k})$, every $\bar{k}$ -point lies in the identity component. Thus

$$\mathrm{SO}'_{n,\bar{k}} = (\mathrm{SO}'_{n,\bar{k}})^\circ,$$

and $\mathrm{SO}'_{n,\bar{k}}$ is connected.

(f) From part (a), for every $R \in \text{Alg}_{\bar{k}}$,

$$O_{n,\bar{k}}(R) \simeq O'_{n,\bar{k}}(R).$$

This isomorphism is given by conjugation, hence preserves determinant. Therefore it restricts functorially to

$$\text{SO}_{n,\bar{k}}(R) \simeq \text{SO}'_{n,\bar{k}}(R),$$

and by Yoneda

$$\text{SO}_{n,\bar{k}} \simeq \text{SO}'_{n,\bar{k}}.$$

By part (e), $\text{SO}'_{n,\bar{k}}$ is connected. Hence SO_n is geometrically connected, and in particular connected. $\square$

Exercise A.8. Let k be algebraically closed.

- (a) Let G be a closed subgroup of GL_n over k . Then $\mathbb{A}^n$ has a natural action of G . Determine the orbits and closed orbits for

$$G = \text{GL}_n, T_n, \text{SL}_n.$$

- (b) The natural action of $G = \text{GL}_2$ on $\mathbb{A}^2$ induces an action of G on $\mathbb{P}^1$ (as $\mathbb{P}^1$ is the quotient of $\mathbb{A}^2 \setminus \{(0,0)\}$). Please determine the orbits for the action of G on $\mathbb{P}^1$ and the diagonal action of G on $\mathbb{P}^1 \times \mathbb{P}^1$. Please also indicate the closed orbits.

Proof. (a) For $G = \text{GL}_n$, the orbits are

$$\mathbb{A}^n \setminus \{0\}, \quad \{0\}.$$

Indeed, GL_n acts transitively on the nonzero vectors. The only closed orbit is $\{0\}$.

For $G = T_n$, the orbits are indexed by subsets

$$S \subseteq \{1, \dots, n\}.$$

For such an S , the corresponding orbit is

$$\mathcal{O}_S = \{(x_1, \dots, x_n) \in \mathbb{A}^n \mid x_i \neq 0 \iff i \in S\}.$$

Equivalently,

$$\mathcal{O}_S \simeq \text{Spm}(k[X_i \mid i \in S]_{\prod_{i \in S} x_i}) \subseteq \text{Spm } k[X_i \mid i \in S] \subseteq \mathbb{A}^n.$$

Its closure is the corresponding coordinate subspace, so the only closed orbit is $\{0\}$.

For $G = \mathrm{SL}_n$, when $n = 1$, the group SL_1 is trivial. Hence every point is an orbit, and every orbit is closed. When $n \geq 2$, SL_n is transitive on $\mathbb{A}^n \setminus \{0\}$: extend a nonzero vector to a basis and rescale one of the remaining basis vectors so that the resulting matrix has determinant 1. Thus the orbits are

$$\mathbb{A}^n \setminus \{0\}, \quad \{0\},$$

and the only closed orbit is $\{0\}$.

(b) By the discussion above, the orbit of $G = \mathrm{GL}_2$ on $\mathbb{P}^1$ is all of

$$\mathbb{P}^1.$$

Hence it is closed.

For the diagonal action of G on $\mathbb{P}^1 \times \mathbb{P}^1$, there are two orbits:

$$\Delta, \quad (\mathbb{P}^1 \times \mathbb{P}^1) \setminus \Delta,$$

where

$$\Delta = \mathrm{im}\left(\mathbb{P}^1 \xrightarrow{\mathrm{diag}} \mathbb{P}^1 \times \mathbb{P}^1\right).$$

Indeed, Δ is one orbit by transitivity on $\mathbb{P}^1$. If $x \neq y$, nonzero representatives of the two distinct lines form a basis of k^2 , so every ordered pair of distinct points belongs to the same orbit. Since $\mathbb{P}^1$ is complete, the diagonal morphism is proper and its image Δ is closed. Therefore Δ is the only closed orbit. $\square$

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